UNIVERSITÀ DEGLI STUDI DI MILANO

FACOLTÀ DI SCIENZE E TECNOLOGIE

Corso di Laurea Magistrale in Informatica

SOCIAL PRESENCE AND EMOTIONAL INTERACTION: COMPARING NAO ROBOT AND ITS DIGITAL TWIN

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Academic Year 2024-2025

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Abstract

The development of social robots offers invaluable insights into human-robot interaction, but their high cost and logistical complexity often limit the scope and accessibility of research. This has fueled a critical debate within the field: can a virtual robot serve as a valid substitute for its physical counterpart, especially when it comes to conveying nuanced social cues like emotion? Answering this question is crucial for enabling more scalable, flexible, and affordable HRI studies.

This thesis addresses this challenge directly. We designed and developed a comprehensive experimental platform to empirically compare human perception of emotional expressions from a physical NAO robot versus its meticulously crafted, kinematically identical digital twin. The core of our work is a high-fidelity virtual model and a unified software architecture capable of dispatching the exact same motion commands to both platforms. This setup ensures that physical embodiment is the sole independent variable, allowing for a rigorous and controlled comparison.

Through a comparative user study, we collected data on how participants recognized a series of non-verbal emotional gestures. Our findings reveal no statistically significant difference in recognition accuracy between the physical and virtual conditions. The legibility of the gesture itself, rather than the robot's physical presence, emerged as the determining factor for successful communication. This result provides strong evidence that, for the specific task of interpreting affective body language, a well-designed digital twin is a reliable and effective research tool.

Furthermore, this work demonstrates the practical utility of our validated virtual platform by integrating it into a novel framework for real-time, empathetic dialogue driven by a Large Language Model (LLM). This extension serves as a proof-of-concept, showcasing how our findings can be used to prototype and refine more complex, multimodal interactions in a virtual environment before deployment on physical hardware.

Chapter 1

Introduction

Human-Robot Interaction (HRI) has rapidly emerged as a significant and multidisciplinary field, bridging computer science, robotics, and psychology. Its core aim is to enable seamless and effective collaboration and communication between humans and robotic systems. Within this expansive domain, social robots represent a particularly compelling area of research. These robots are specifically designed to interact with humans in a socially intelligent and engaging manner, fulfilling roles in education, healthcare, companionship, and customer service.

Within this field, the NAO humanoid robot, developed by SoftBank Robotics, has become a standard and widely adopted platform. A comprehensive review of a decade of research confirms its broad use in HRI studies, particularly for its approachable design and expressive capabilities, which make it suitable for interactions with diverse user groups, including children and the elderly [1]. However, while the deployment of physical social robots offers invaluable insights into embodied interaction, they often entail significant costs and logistical complexities, potentially limiting the scope and accessibility of research. This presents a practical challenge for many researchers and institutions. Consequently, there is a growing need to explore alternative methodologies that can facilitate robust HRI studies without the immediate necessity of expensive physical hardware. This has led to a critical and unresolved debate within the field: can a virtual robotic counterpart serve as a valid substitute for a physical one? On one hand, virtual agents, whether rendered on a screen or experienced in immersive virtual reality, offer a scalable, cost-effective, and flexible alternative, enabling rapid prototyping and iteration without the limitations of physical hardware. On the other hand, a compelling body of work suggests that physical embodiment—the tangible co-presence of a robot in a shared physical space—is crucial for fostering genuine social connection.

For instance, some studies have shown that interactions with virtual robots can be perceived as more ”discomforting” and that people maintain a larger personal distance from them compared to physical robots [2]. Conversely, other research, including a recent meta-analysis, suggests that for certain tasks and under specific conditions, there are no significant differences in user perception of social presence or engagement between physical and virtual agents [3, 4]. This indicates that a well-designed virtual agent can, in some scenarios, be an effective substitute. This ongoing debate reveals a significant gap in the literature. While factors like proxemics and social influence have been compared, a focused, quantitative investigation into how physical versus virtual embodiment specifically affects the human perception of a robot's non-verbal, emotional cues remains largely unexplored. This is particularly relevant for robots like NAO, which rely on bodily gestures rather than complex facial mechanics to convey emotion.

To address this gap, this thesis aims to provide an empirical answer to a central question: Does the physical presence of a social robot fundamentally alter how humans perceive its emotional expressions compared to a visually and kinematically identical digital twin?

To systematically investigate this question, a rigorous methodology was developed to empirically compare user interactions across both embodiments. The cornerstone of this approach was the creation of a high-fidelity digital twin of the NAO robot, engineered to achieve visual and kinematic parity. This was not merely an animation, but a fully articulable virtual model with an identical skeletal structure and degrees of freedom, ensuring its movements precisely mirrored the timing, trajectory, and subtleties of the physical robot's actions. A curated set of 12 emotional gestures, derived from established HRI literature and representing multiple variations of four basic emotions (happiness, sadness, anger, and fear), was implemented to provide robust and non-repetitive stimuli. Furthermore, a unified software architecture was engineered to dispatch the exact same motion commands to both the physical and virtual platforms. This control system was the key to the experimental design, as it guaranteed that embodiment was the sole independent variable under investigation.

The work presented in this thesis is therefore guided by two interconnected objectives: The primary objective was to empirically investigate the influence of physical embodiment on emotion recognition. This involved a direct, quantitative comparison of how accurately participants could identify emotional gestures performed by the physical NAO robot versus its high-fidelity digital twin. The goal was to provide clear data to the ongoing embodiment debate and determine if a virtual counterpart can serve as a valid proxy for its physical version in studies of affective communication.

The secondary objective was to design and implement a technical framework driven by an empathetic Large Language Model (LLM). This framework is capable of driving real-time, emotionally expressive dialogue on both the physical NAO and its digital twin. It serves as a forward-looking application of our findings, demonstrating how a validated virtual platform can be used for rapid prototyping and refinement of complex, multimodal interactions. By engineering this system to operate seamlessly across both platforms, this work lays the groundwork for future studies investigating the role of embodiment in richer, more dynamic social contexts, moving beyond static gestures to integrated verbal and non-verbal communication.

This thesis is structured as follows:

- Chapter 2 provides a review of the state of the art, covering the NAO robot's role in social HRI, the ongoing academic debate on physical versus virtual embodiment, and the theoretical models of emotion relevant to this work.
- Chapter 3 details the methodology, including the technical development of the high-fidelity digital twin, the unified software architecture, and the design of the comparative user experiment.
- Chapter 4 presents the quantitative results of the study, focusing on emotion recognition accuracy and dimensional analysis.
- Chapter 5 provides an in-depth discussion and interpretation of these results, contextualizing them within the existing literature and exploring their theoretical implications.
- Chapter 6 summarizes the key conclusions of the thesis and outlines promising directions for future research.

Chapter 2

State Of The Art

The field of Human-Robot Interaction (HRI) is a rapidly evolving discipline focused on creating effective and natural collaboration between humans and robots. A key area within HRI is the development of social robots, which are designed to engage with people in a socially intelligent manner. These robots are increasingly being explored for roles in education, healthcare, and companionship. Among social robots, NAO has earned particular recognition in scientific research. This chapter reviews the foundational literature that informs the present work, beginning with the NAO robot's role as a social platform, moving to the critical comparison of real versus virtual interaction, discussing the theoretical basis for emotion recognition especially in relation to NAO and concluding with the motivation for this thesis.

2.1 NAO as a Social Interaction Robot

The NAO humanoid robot, developed by SoftBank Robotics, is one of the most widely used platforms in human-robot interaction (HRI) research. Its compact size, 25 degrees of freedom, and approachable, friendly design make it an effective social agent capable of engaging users across different age groups, including vulnerable populations such as children and the elderly. NAO's expressive range of motion combined with multimodal communication channels (verbal speech, gestures and changes in eye color) enable it to convey social cues that facilitate interaction.

Primarily, NAO functions in roles such as peer, tutor, or mediator, frequently applied in educational contexts and therapeutic settings. In particular, it has been used extensively to support children with Autism Spectrum Disorder (ASD), where its predictable and simplified behavior can help reduce social anxiety and foster the development of communication and social skills.

A comprehensive scoping review spanning a decade of research [1] confirms NAO's broad adoption in short-term, dyadic human-robot interactions. The robot's strengths include its friendly appearance, ease of programming and deployment, and versatility across diverse domains such as education, therapy, eldercare, and even artistic performances. Despite these advantages, NAO faces several limitations. Its speech recognition system performs optimally only in quiet environments and close proximity, limiting natural verbal interaction, especially with children. Mobility is constrained to flat, hard surfaces, and extended autonomy remains a challenge, with many studies relying on "Wizard of Oz" control paradigms where a human operator guides the robot's behavior.

More recently, a significant trend in HRI is the integration of Large Language Models (LLMs) to enhance the social capabilities of robots like NAO. This integration aims to move beyond pre-programmed, scripted dialogues towards more fluid, natural, and context-aware conversations. By leveraging the power of LLMs, a social robot can generate empathetic and relevant responses in real-time, fundamentally transforming the quality of interaction and paving the way for more sophisticated and meaningful human-robot relationships. Billing et al. [5] present the first integration of an LLM with NAO and Pepper robots, demonstrating that such models can significantly improve conversational fluency, contextual relevance, and user engagement. Their findings suggest that LLM-driven interactions are perceived as more natural and flexible compared to scripted approaches, highlighting the potential of this technology to support more autonomous and socially intelligent robotic behavior. For instance, a recent study [6] demonstrates an application connecting a Pepper robot, a platform similar to NAO and produced by the same company, to ChatGPT to assist individuals with Autism Spectrum Disorder. The work highlights how LLM integration can make interactions more natural, personalized, and effective.

2.2 Physical vs Virtual Embodiment in HRI

While physical robots provide useful data on embodied interaction, their deployment is often hindered by significant cost, logistical complexity, and maintenance requirements. This has motivated a growing body of research into virtual counterparts, which offer a more accessible and scalable alternative for HRI studies. These virtual environments can range from screen-based simulations to immersive Virtual Reality (VR) and Augmented Reality (AR) experiences.

The assumption that a virtual robot can serve as a perfect substitute for a physical one is a subject of ongoing investigation. A critical study by Li et al. explored the differences in human-robot proxemics between the real world and VR [2]. Their findings revealed that participants maintained a significantly larger personal distance from the virtual robot compared to the physical one. Furthermore, the virtual robot was perceived as more "discomforting". These results suggest that the absence of physical presence, coupled with potential limitations of VR systems, can fundamentally alter a user's perception and behaviour. This perceptual gap challenges the direct transferability of findings from virtual to real-world settings and underscores the need for direct comparative studies.

Many studies find the physical NAO robot more effective than its virtual version, mainly due to its tangible embodiment. However, recent work by Thellman et al. [3] directly compared a physical NAO robot to its virtual screen-based counterpart using the Ultimatum Game paradigm. Contrary to expectations, the study found no significant differences in perceived social presence or social influence between the physical and virtual robots. Interestingly, it was not the physical embodiment per se, but rather the perceived social presence of the agent that significantly influenced participants' responses. Participants who rated the robot, regardless of embodiment, as more socially present were more likely to accept its offers. This finding suggests that the effectiveness of an agent in social interaction may depend less on physical instantiation and more on its ability to be experienced as a socially capable actor. Although the study aligns with Li's conclusion that physical presence plays a role in social interaction, Thellman et al. suggest that it is actually the agent's perceived social presence—regardless of physical or virtual embodiment—that most strongly influences social outcomes, prompting a reevaluation of how and when virtual agents can serve as adequate proxies for physical robots in social HRI scenarios.

Recent meta-analytic evidence by Esterwood et al. (2025) [4] further supports these findings, showing no significant differences in anthropomorphism, social presence, or engagement between physically collocated and non-physically collocated robots. Their results emphasize the importance of contextual factors and functional embodiment over mere physical presence, suggesting that virtual robots can effectively substitute physical ones in many HRI scenarios, depending on the task and user characteristics.

2.3 Modelling and Recognizing Emotions with NAO

To quantitatively assess the difference between real and virtual interactions, a robust metric is required. Emotion recognition serves as an excellent candidate, as the ability to accurately perceive and interpret emotional expressions is fundamental to social interaction. To date, no comprehensive study has quantitatively compared human recognition of emotional expressions between a real robot and its digital twin, making this an important gap in the literature.

2.3.1 Theoretical Models of Emotion

The study of emotional expression and recognition in Human-Robot Interaction (HRI) is often grounded in established psychological theories of emotion. Among the most influential is Paul Ekman's model of basic emotions. Ekman's pioneering research in the 1960s and 1970s, conducted through extensive cross-cultural studies, identified six universal emotions—happiness, sadness, anger, fear, disgust, and surprise—that are consistently recognized across diverse human populations [7]. He demonstrated that these emotions correspond to distinct, innate facial muscle movements, suggesting a biological basis for emotional expression that transcends cultural differences. Ekman's work not only advanced the understanding of nonverbal communication but also provided a discrete and practical categorization of emotions widely adopted in various fields, including affective computing and HRI.

Complementing this discrete view, James Russell proposed the Circumplex model of affect [8], which conceptualizes emotions along two continuous and orthogonal dimensions: valence, which ranges from pleasant to unpleasant, and arousal, which ranges from high to low activation. Unlike categorical models, the dimensional framework proposed by Russell captures the gradations and blends of emotional experience, accommodating the fluid and often ambiguous nature of human affect. This model has been influential in providing a flexible structure for modeling emotional states that vary in intensity and quality, allowing for richer representations of affective states beyond fixed categories.

Both Ekman's and Russell's models have significantly influenced the design, modeling and interpretation of emotional expressions in HRI. Ekman's basic emotions offer clear, recognizable categories that facilitate the development of expressive behaviors in social robots, enabling straightforward mapping between emotional states and observable robot behaviors such as gestures, facial expressions, and vocal modulations. This discrete framework supports robust emotion recognition by humans, leveraging fundamental and consistent emotional signals that are deeply embedded in human social communication. On the other hand, Russell's circumplex model supports more nuanced and dynamic representations of affective states, which is particularly valuable for robotic systems aiming to exhibit or interpret complex or mixed emotions. This dimensional approach allows robots to express varying intensities of emotions and to smoothly transition between emotional states, enhancing the naturalness and adaptability of interactions.

Discrete primary emotions can be mapped in the continuous Circumplex model. Figure 1 visually maps Ekman's basic emotions onto Russell's valence-arousal space, illustrating how each category corresponds to a specific emotional quadrant.

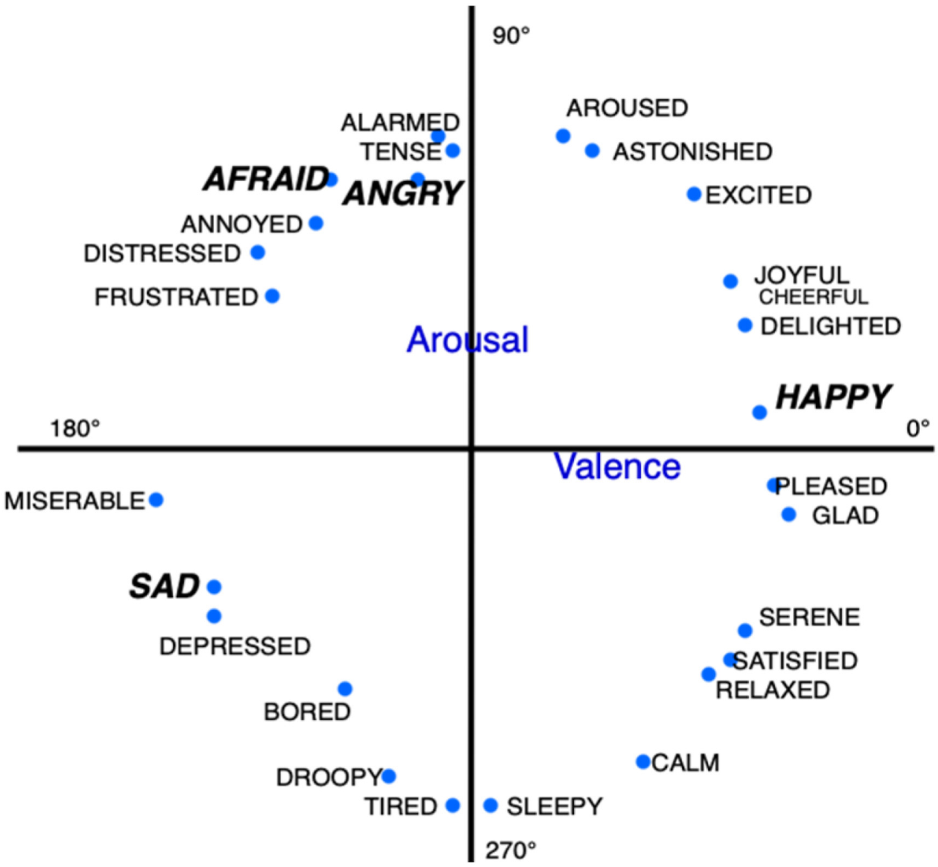

Figure 1: Representation of emotions inside the 2D space of Valence and Arousal [8], where Valence describes the extent to which an emotion is positive or negative, and Arousal refers to its intensity. The primary emotions by Ekman of happiness, sadness fear and anger are labeled in capital letters.

2.3.2 Implementing Emotions on NAO

Researchers have dedicated significant effort to enabling the NAO robot to express emotions, a crucial step for creating natural and believable social interactions. Since NAO lacks complex facial mechanics, its emotional expressions are conveyed multimodally, primarily through a combination of body language (postures and gestures), changes in its LED eye color, and sound. The implementation of these emotional cues is often grounded in psychological research to ensure they are recognizable and interpretable by humans.

A common approach involves creating a predefined “emotional dictionary” where specific combinations of movements, colors, and sounds are mapped to target emotions. For example, Pugi et al. [9] developed an emotional dictionary for NAO to be used in therapeutic games with children with Autism Spectrum Disorder (ASD). They integrated gestures, LED eye colors (e.g., yellow/orange for positive emotions, blue/violet for negative), and speech to convey emotional states. The gestures themselves were refined through a co-design process involving psychologists and therapists to ensure their social appropriateness.

The design of NAO's emotional body language often draws inspiration from studies of human nonverbal cues. Several studies have focused on adapting postures known to convey specific emotions in humans for the NAO's morphology. Erden [10], for instance, adapted quantitative descriptions of human emotional postures (anger, sadness, and happiness) from Coulson's [11] research to the NAO platform. This process required careful transformation to account for the robot's different joint structure, range of motion, and mass distribution. The resulting postures were then validated through user studies to confirm that they successfully conveyed the intended emotions. Similarly, Beck et al. [12] demonstrated that both adults and children could recognize key emotional poses displayed by NAO (such as pride, sadness, and anger) and that subtle changes, like the position of the head, could significantly alter the perceived emotion.

The use of multiple modalities is a recurring theme. Häring et al. [13] created and evaluated expressions for anger, sadness, fear, and joy by combining body movement, sound, and eye color. Their evaluation revealed that body movements were the most reliable single cue for conveying emotion, while eye color was often found to be unreliable, with the exception of red being associated with anger. This highlights the complexity of creating a universally understood mapping for non-biomimetic cues like colored lights. Sounds, particularly non-linguistic vocalizations like crying or cheering, were found to be effective, especially when synchronized with corresponding body movements.

The validation of these implemented emotions is a critical final step. Researchers typically conduct studies in which participants look at the robot's expression and are asked to identify the emotion from a list of options, often based on Ekman's six basic emotions [7]. These studies help to refine the expressions and identify which cues or combinations of cues are most effective at conveying a specific emotional state. Through such iterative processes of design and evaluation, researchers are progressively building a more robust understanding of how a humanoid robot like NAO can communicate its internal state, paving the way for more nuanced and effective human-robot interaction.

2.4 Motivation

The preceding review of the state of the art establishes two fundamental realities: first, that the NAO robot is a highly capable and validated platform for social HRI research, particularly for conveying emotion through multimodal behavior; and second, that a significant and unresolved debate persists regarding the perceptual equivalence of physical robots and their virtual counterparts. The literature presents conflicting evidence: some studies find that physical presence is crucial, enhancing perceptions of trustworthiness and social weight [2]. In contrast, other studies suggest that for certain social tasks, a well-designed virtual agent can be just as effective, and that perceived social presence may matter more than physical form [3, 4].

Within this context of debate, a critical gap emerges. While factors like proxemics, influence, and trust have been compared, a focused, quantitative investigation into how physical embodiment directly affects the human recognition of emotional expressions remains largely unexplored. This is particularly true for robots like NAO, which communicate emotion primarily through bodily posture and gesture rather than complex facial mechanics. This leads to the central research question of this thesis: Does the physical presence of the robot, its tangible occupation of real space, enhance or alter the clarity and perceived intensity of its emotional signals compared to a visually identical, high-fidelity digital twin? Answering this question is not merely a theoretical exercise; it has profound practical implications for the future of HRI, determining whether less expensive and more scalable virtual simulations can serve as valid proxies for physical robot experiments in the affective domain.

To address this gap, this thesis pursues two interconnected objectives that bridge fundamental research with forward-looking application:

1. **To empirically investigate whether the physical embodiment of the NAO robot influences the accuracy and perception of its emotional expressions compared to a visually identical virtual counterpart.** This is the core scientific objective of this work. It will be addressed through a controlled, comparative user study in which participants classify emotions expressed by both the real and virtual NAO, providing quantitative data to contribute directly to the embodiment debate.

2. **To design and implement a framework for an empathetic Large Language Model (LLM) capable of driving real-time, emotionally expressive dialogue on NAO.** This secondary objective serves as a key technical contribution. It contextualizes our primary investigation by demonstrating a practical, next-generation application for our findings. The framework is engineered to operate across both the physical and virtual platforms, laying the groundwork for advanced interactive systems and underscoring the importance of understanding any perceptual differences between the two modalities.

Furthermore, while previous research provides valuable insights, many comparisons have relied on video recordings or basic animations, which may introduce confounding variables such as graphical style, differing contexts, or a lack of real-time interactivity. These limitations can make it difficult to isolate the true effect of physical presence.

To overcome these methodological hurdles, this thesis leverages a meticulously crafted high-fidelity digital twin: a virtual model that is not merely an animation, but a functional and visual replica of the physical NAO. This approach ensures that the only significant variable under investigation is the physical embodiment itself, allowing for a more rigorous and valid comparison.

Chapter 3

Method

This chapter details the methodology employed to investigate the research questions outlined in the previous chapter. The methodology is presented in two main parts. The first section, System Implementation and Design, describes the technical development of the experimental platform, including the creation of the emotional stimuli, the design of the high-fidelity digital twin, and the software architecture that controls both the physical and virtual robots. The second section, Experimental Procedure, details the design of the user study, the implementation of emotional gestures in the experiment, and the step-by-step process used to collect data for comparing the two conditions.

3.1 System Implementation and Design

A robust and controlled experimental setup is crucial for a valid comparison between a physical robot and its virtual counterpart. This section outlines the functional requirements that guided the development and the technical architecture of the resulting system.

3.1.1 Functional Specifications

In order to rigorously compare the two conditions and ensure that physical embodiment was the only systematically varied factor, the experimental system was developed according to two essential functional principles.

- **High Fidelity and Parity:** The virtual NAO robot must be a high-fidelity digital twin of the physical robot. This implies strict parity across multiple dimensions:
  - **Visual Fidelity:** The virtual robot's appearance and materials must be a 1:1 replica of the physical model. The virtual environment must also mirror

the real-world experimental setting to minimize contextual differences.

  - **Kinematic and Behavioral Fidelity:** The virtual robot must possess the same 25 Degrees of Freedom (DoF) as the physical robot. Crucially, both robots must be driven by the exact same motion commands, ensuring that all gestures, postures, and animations are kinematically identical in terms of joint rotation, velocity, and timing.

- **A Standardized and Recognizable Emotion Set:** To create a valid test of emotion recognition, the emotional expressions used must be grounded in established research. Based on the literature reviewed in Chapter 2, a discrete set of basic emotions was chosen. For each emotion, multiple distinct gestures were implemented to allow for a pseudo-randomized presentation during the experiment, preventing participants from merely memorizing a single animation per emotion. Four basic emotions—happiness, sadness, anger, and fear—were selected, and three distinct gestures were designed for each, resulting in a total of 12 unique emotional stimuli.

3.1.2 High-Fidelity Digital Twin Design

The creation of the digital twin and the emotional stimuli followed a systematic pipeline to ensure the functional specifications were met.

Virtual Robot Development Pipeline

Prior to the actual implementation of the virtual robot, an extensive search for existing models was conducted. However, many of them turned out to be too rudimentary and not faithful replicas of the NAO robot. In addition, while it is possible to use virtual NAO simulators —such as the native one integrated into Choregraphe, the official NAO simulator released by the parent company— these come with significant limitations, including the inability to animate the eyes color and movement, as well as the difficulty of integrating the robot model with custom code for smooth, real-time interaction.

The creation of the virtual NAO followed a step-by-step 3D development pipeline to ensure a 1:1 representation of the real robot, this custom implementation ensures that the virtual model is not just playing a pre-rendered video, but is a functional, articulable entity that perfectly mirrors the physical robot's movements. This modular design makes the digital twin reusable for any project that requires a virtual NAO controlled by Choregraphe animations.

1. **Rigging and Model Preparation:** A 3D model of the NAO robot, based on the original CAD files available online, was imported into Blender, an open-source 3D modeling software. In Blender, a skeleton (or "rig") was created to match

NAO's 25 joints and applied to the imported mesh. Each part of the mesh was carefully weighted to move appropriately with its associated joint.

The mesh was then modified to reflect all the visual features of the robot: each component was separated to allow for the assignment of different materials, and the eyes were divided into eight parts each so that they could later be animated via code in Unity.

Finally, materials were applied to the mesh to replicate the plastic and metallic finishes of the physical robot.

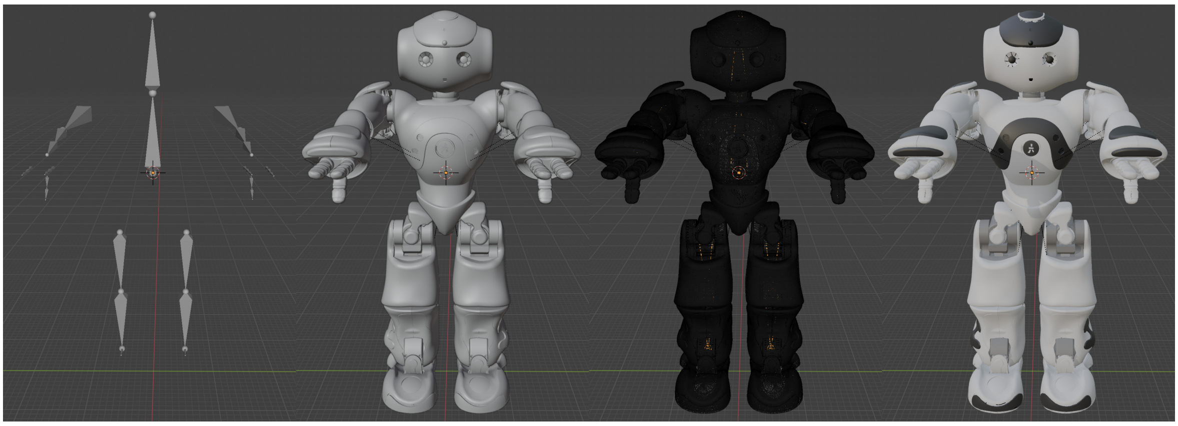

Figure 2: 3D robot model implementation process in Blender.

2. **Unity Integration and Kinematic Mapping:** The model was then imported into the Unity game engine. This is one of the most critical steps for ensuring fidelity. A `C#` script was developed in Unity to act as both a parser and a controller. This script reads the keyframe animations data exported from Choregraphe and maps them in real time to the corresponding joints of the virtual robot's rig. The Python movement code from Choregraphe is exported and saved as a plain text file (e.g., movement1.txt). A keyframe extraction routine then reads this file and executes the animation accordingly. A precise 1:1 mapping between the coordinate systems and joint names of the physical NAO robot and the Unity rig was established. This was necessary because the NAO robot defines joint orientations individually in terms of roll, pitch, or yaw and often not all three simultaneously, as illustrated in Figure 3. In addition, an interpolation function was implemented to transition each joint smoothly from one position to another, following the timing defined by the keyframes. This ensures that the robot's movements are fluid, following a realistic temporal distribution between keyframes, and faithfully replicating the motion of the real robot. It is important to note that this application is designed to replicate the real robot's motion on

the virtual model. However, it would not be complex to reverse this process, using motion applied to the virtual robot in Unity to control the real robot. This opens new possibilities for developing and testing real-world motions within a simulated environment, and then applying them to the real robot.

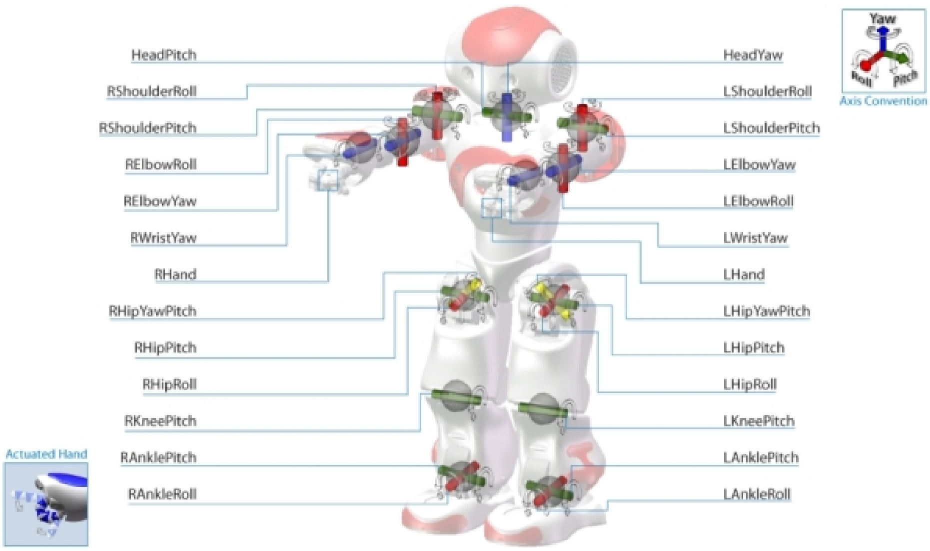

Figure 3: Composition and orientation of the NAO robot joints.

Environmental and Visual Parity

To fulfill the functional requirement of high-fidelity parity, it was not enough to simply create an accurate 3D model. The virtual environment in which the digital twin was presented had to be a meticulous replica of the physical experimental setting. This step was crucial for minimizing confounding variables and ensuring that participants' perceptions were influenced by the robot's embodiment rather than by differences in background, lighting, or camera perspective—as can occur with some simulators, such as the one integrated into Choregraphe, where the background is entirely blue.

- **Background and Staging:** The virtual robot was positioned on a plain dark grey surface, identical to the table used in the physical setup. The background was rendered as a neutral office environment, with color tones and assets chosen to resemble the real-world setting where the experiment took place.
- **Lighting:** A single directional light was configured in Unity to replicate the primary overhead light source from the physical room's ceiling. This was essential

to ensure that the virtual robot's plastic and metallic materials appeared visually consistent with those of the real robot.

- **Camera Perspective:** The virtual camera in Unity was positioned and configured so that the field of view and camera angle matched the real-world setup, ensuring that the framing of the virtual robot on the monitor appeared identical to that of the real robot when a participant stood in front of the table at both stations.

The result of this process, as illustrated in Figure 4, is a virtual condition that is visually and contextually nearly identical to the physical one, thereby isolating physical presence as the primary experimental variable.

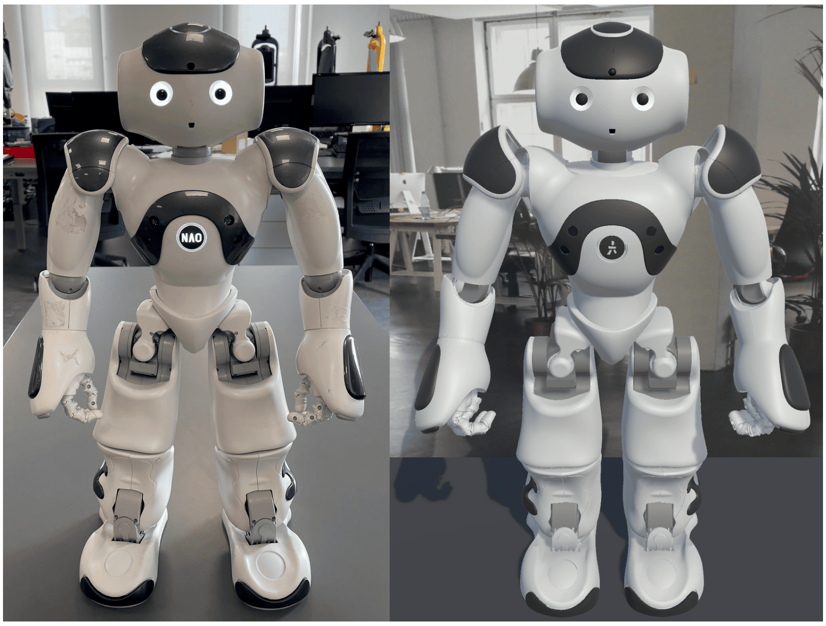

Figure 4: Side-by-side comparison of the physical NAO robot (left) and its high-fidelity digital twin in Unity (right).

3.1.3 Emotion Stimuli Selection and Creation

The 12 emotional gestures were designed using SoftBank Robotics' proprietary Choregraphe software. Each gesture was implemented as a keyframe animation, defining a sequence of joint positions over time. This approach allowed participants to observe not just a static pose, but a dynamic animation conveying the intended emotional expression. The design process was informed by existing literature on emotional body language in HRI, particularly studies that validated emotional gestures and postures specifically for the NAO platform.

As discussed in Chapter 2, Ekman's theory identifies six basic emotions. For this study, only four of these were selected. Disgust was excluded because it is predominantly conveyed through facial expressions, which the NAO robot cannot reproduce effectively. Surprise was also omitted, as it is considered a complex emotion that can carry both positive and negative valence, complicating its interpretation in experimental settings [14, 15].

Limiting the set to only one expression per basic emotion—such as a single gesture for happiness, sadness, anger, and fear—was deemed insufficient for the aims of this study. These four emotions, while distinct and easily identifiable, offer limited variability. A broader set of emotional gestures was therefore necessary to ensure a more diverse and balanced distribution of stimuli, enabling pseudo-random presentation and minimizing response predictability.

The selection of emotions was guided primarily by the frequency of use in prior studies involving the NAO robot. When this was not possible due to the unavailability of certain emotions, the search was extended to studies involving other agents, focusing on emotions associated with recognizable human postures within the broader field of HRI.

The final set of 12 gestures is summarized in Table 1 and shown in Figure 5.

Paper	**Ha1**	**Ha2**	**Ha3**	**Sa1**	**Sa2**	**Sa3**	**An1**	**An2**	**An3**	**Fe1**	**Fe2**	**Fe3**
Häring [13]			X		X	X		-		X		
Beck [12]	X		-	X			-			X		
Cohen [16]	X			X			X					X
Erden [10]	X			X					X			
English [17]	X				X	X	X					X
Miskam [18]	X			X			X					X
Alimardani [19]	X					X	X			X	X	
Valenti [20]	X	X		X	X							
Greczek [21]	X			X			X			X		-
Consoli [22]			X	X			X			X		

Table 1: Choice of emotions based on the robot NAO literature. "X" means that the emotion is present, "-" that a similar version is present.

Although "Anger2" has not been reported in studies that use the NAO robot, it is a widely used emotion in HRI. Indeed, it has already been implemented on other humanoid platforms, as shown in earlier research by Nishiwaki et al. [23], Mazzoni et al. [24], and Chitti et al. [25].

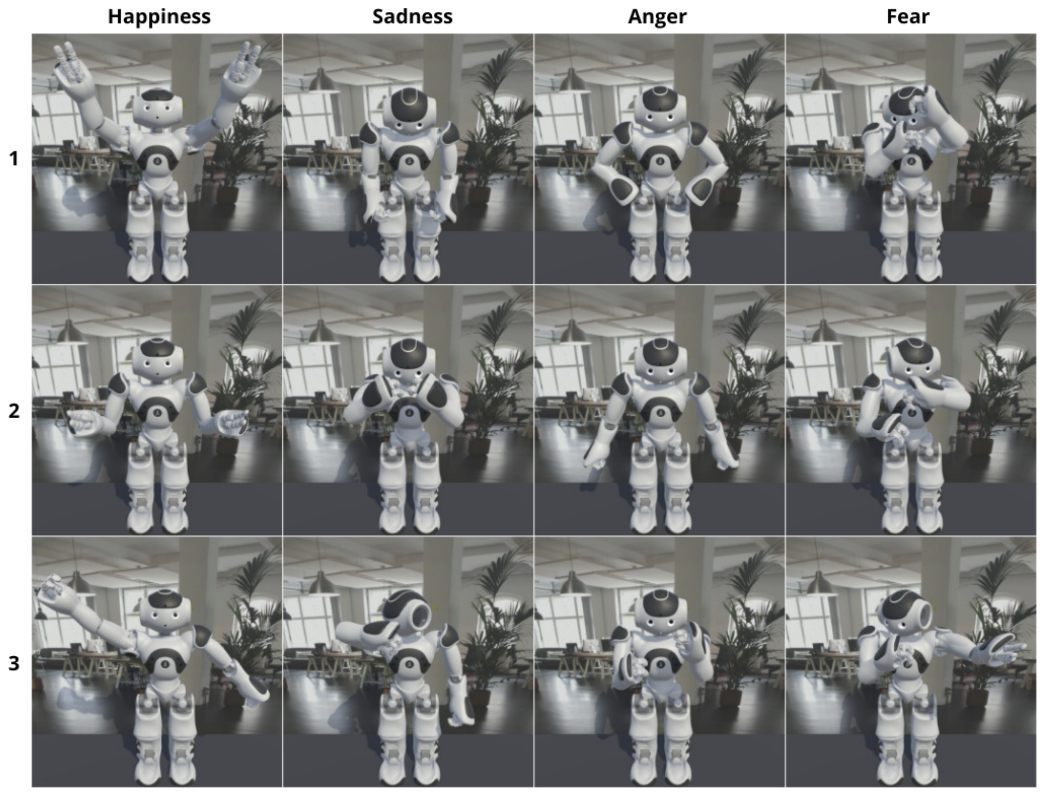

Figure 5: Emotion gestures: Happiness, Sadness, Anger, Fear. Three gestures for each emotion. Look at Figure 10 for a direct comparison with the physical robot's gestures.

3.1.4 Framework Architecture and Realization

A unified software framework was developed to control both the experiment flow and the robot's actions, whether physical or virtual. The architecture is designed to ensure that the core experimental logic is completely decoupled from the robot's embodiment, thereby isolating physical presence as the primary independent variable. The system is based on a client-server model, with a central Python application, built using the Flask web framework, acting as the server and orchestrator for the entire experiment.

Central Orchestrator and Experimental Logic

The core of the system is a Flask web server (`main-ob0.py`) responsible for managing the entire experimental procedure. Its primary functions include:

- **User Interface:** Serving the web-based user interface (`questions.html`) through which participants submit their responses.

- **Stimulus Management:** Randomizing the presentation order of the 12 emotional gestures and counterbalancing the conditions (physical vs. virtual) across participants to prevent ordering effects.
- **Data Logging:** Receiving participant responses—including the forced-choice emotion classification and the 9-point SAM ratings for valence and arousal—and saving them to a unique, timestamped JSON file for each session.

This centralized approach ensures that every participant follows the exact same procedure, regardless of which robot condition they are assigned to first.

Dual Control Pathways

From the central orchestrator, two distinct pathways are used to control the physical and virtual robots. Crucially, the orchestrator sends the exact same command string (e.g., `‘‘Happiness1’’`) to trigger an action, ensuring behavioral parity.

1. **Physical Robot Control:** The physical NAO robot is controlled via the proprietary `pynaoqi` library. Due to library constraints requiring a Python 2.7 environment, a dedicated Flask-based server script (`main-ob0-gestures.py`) acts as a bridge.
   - When a gesture is to be performed, the main orchestrator sends an HTTP POST request containing the gesture name to this bridge server.
   - The bridge server receives the command and uses the `pynaoqi` library to call the corresponding gesture on the physical NAO robot.
   - Each gesture is a Python script exported directly from the Choregraphe software suite. This ensures that the joint trajectories, timing, and kinematics are a faithful execution of the original design.
2. **Virtual Robot Control:** The virtual robot, rendered in the Unity 3D engine, is controlled via a direct TCP socket connection.
   - The main orchestrator transmits the gesture name via a TCP socket to the Unity application, where it is received by the `C#` script (`SocketServer.cs`).
   - Upon receipt, it calls the `PlayMotion` coroutine in the `NaoMovements.cs` script. This script reads from the text file containing the keyframe data for the specified gesture. This data (joint names, rotation values, and timestamps) is also exported from the same Choregraphe project used for the physical robot.

- The `NaoMovements.cs` script then interpolates these keyframes in real-time, applying the rotations to the corresponding joints of the high-fidelity 3D model.

This dual-pathway architecture, originating from a single command source and utilizing identical keyframe data, guarantees that the only significant difference between the two conditions is the physical versus virtual embodiment of the robot.

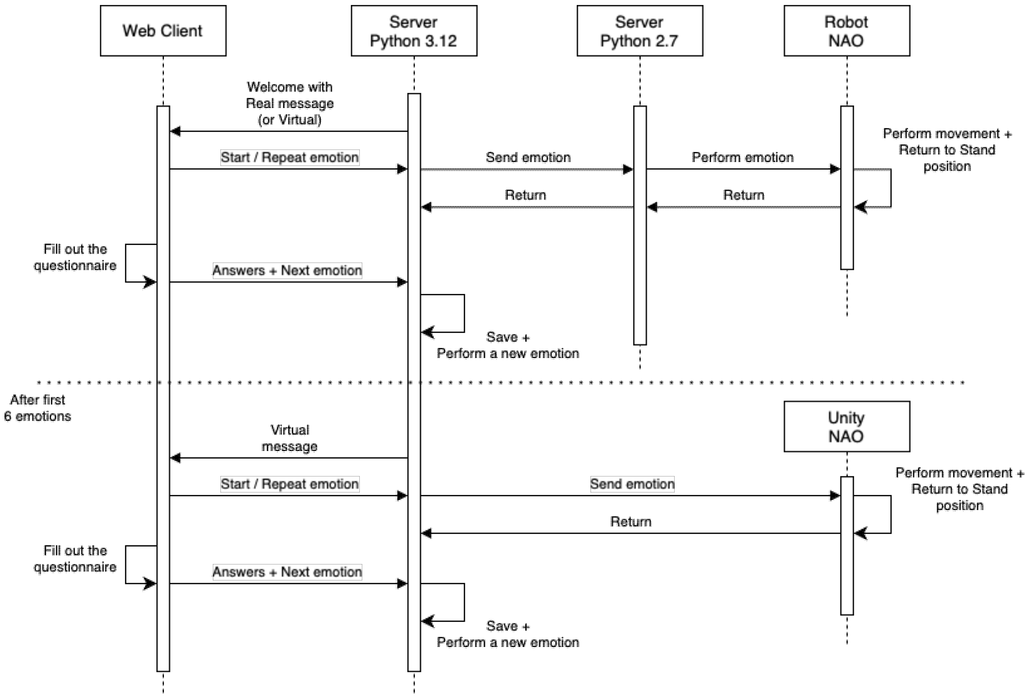

Figure 6: Sequence diagram of the base experimental framework. The main server orchestrates the experiment, sending identical commands to either the physical robot controller or the virtual robot controller depending on the condition.

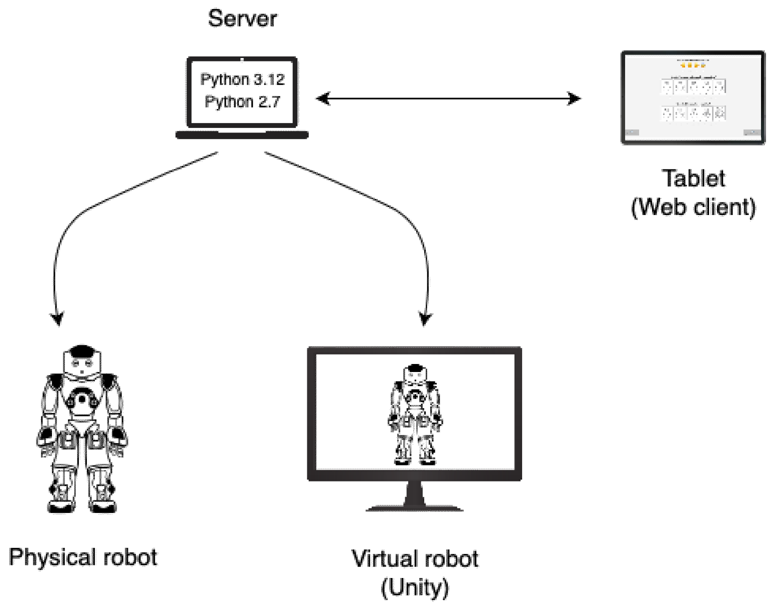

Figure 7: 3D visualization of the experimental framework architecture.

3.1.5 Integration of the Empathetic LLM Framework

As a secondary objective, and to serve as a proof-of-concept for future work, the base framework was extended to support real-time, empathetic dialogue driven by a Large Language Model (LLM). This required a more complex architecture to handle audio input, processing, and synchronized verbal and non-verbal output on both platforms.

The core of this system is Google's Gemini 2.5 Flash model, selected for its multimodal capabilities and accessed through its official API. A crucial aspect of the integration lies in the system prompt provided to Gemini, which guides the model to assume the persona of the NAO robot and embed emotional markers—such as `[happy]`, `[sad]`, `[angry]`, and `[fear]`—into its responses. These tags are used to modulate the robot's expressive behavior accordingly. The specific prompt used alongside the user's input is as follows:

> *"You are the NAO robot. Answer the question naturally and never repeat the other person's words. Use the vocal tokens [happy], [sad], [angry], [fear] at the beginning of sentences or words to change your tone of voice. Use the token [rst] when you want to return to a neutral tone. Maintain a mostly neutral tone and use the Russell model to manage the five emotions."*

This design is intentionally flexible and adaptable to a wide range of contexts. By simply modifying the system prompt, the robot's communication style can be tailored to specific needs. For example, adding the instruction:

> *"Always respond in an empathetic and reassuring tone, adapting to the other person's emotional state to make them feel understood and at ease."*

would shift NAO's behavior toward a consistently empathetic and comforting tone. This modularity makes the framework highly reusable across diverse interaction scenarios.

Dialogue Flow with the Physical NAO

1. **Audio Input:** A dedicated server (`serverNAO.py`), running on the same local network as the NAO robot, uses the `ALAudioDevice` proxy from the `NAOqi` framework to capture a live audio stream from NAO's microphones. The continuous nature of this stream offers several key advantages. Firstly, it allows for more flexible management of interruptions—for example, stopping the audio capture as soon as it is detected that the user has finished speaking, or in response to recording errors. More importantly, since the audio is streamed in real time, it is not stored locally on the NAO robot but is immediately available on the server. This enables direct and immediate forwarding of the audio data to Gemini, as will be explained in a later section.

2. **Processing and LLM Interaction:** The audio stream is forwarded to the main LLM controller (`serverLLM.py`). This script saves the captured speech as an audio file (`.ogg`), a format that offers efficient compression while maintaining high audio quality, a crucial factor for minimizing latency when uploading the audio to the Gemini API. This audio file, rather than just its text transcription, is sent to Gemini. This approach allows the model to analyze not only the words but also the prosodic cues (tone, pitch, pace) in the user's voice, enabling a more contextually empathetic response.

3. **Output Generation and Execution:** Gemini returns a text response containing both conversational content and the embedded emotional tags. The proxy's built-in text-to-speech (TTS) engine vocalizes the response, while the tags are used to modulate the voice's prosody (e.g., `[happy]` is mapped to `vct=105 rspd=105`). The tags are also used to perform one of the 12 emotions mentioned above based on the received tag.

Dialogue Flow with the Virtual NAO

The flow for the virtual twin is architecturally parallel but adapted for a standard desktop environment.

1. **Audio Input:** The LLM controller for the virtual robot (`main-unity.py`) uses the `speech_recognition` library to capture audio directly from the computer's default microphone.

2. **Processing and LLM Interaction:** As with the physical robot, the captured audio is saved as a file and sent to the Gemini API for analysis and response generation. The core interaction with the LLM is identical.

3. **Output Generation and Execution:** This stage demonstrates a more decoupled approach to expression:

   - **Gestures:** The `speak_and_send_tags` function parses the LLM's response. When an emotional tag like `[happy]` is encountered, it immediately sends a command (e.g., `''Happiness1''`) via TCP socket to the Unity application to execute a specific, pre-designed emotional gesture.
   - **Speech:** Simultaneously, the textual part of the response is sent to the physical NAO's TTS engine, which renders it into an audio file. This file is then transferred back to the local machine and played through the computer's speakers using the `playsound` library. This approach ensures that the virtual robot speaks with the same distinctive NAO voice as the physical one.

     It is important to note that the physical NAO robot is not required for this process to function. The decision to rely on the NAO's native TTS engine was made to guarantee consistency in vocal identity between the two embodiments, as locally tested TTS solutions were unable to convincingly replicate NAO's characteristic voice. However, if only the virtual robot is used, any local TTS engine—such as the one included but commented out in the original code—can be employed instead.

In the interaction with NAO, the dialogue can be segmented into distinct states. These include: the user speaking, the transmission of the audio, the generation of the response by the Gemini model, and finally NAO's text-to-speech (TTS) output accompanied by related animations. Each of these states is associated with a specific eye color animation on NAO, designed to provide intuitive feedback and guide user behavior during the interaction.

When it is the user's turn to speak, NAO's eyes display a green color, indicating that the robot is actively listening. Upon completion of the user's utterance, NAO nods its head to signal understanding. Simultaneously, its eyes shift to a blue "loading" animation, representing the processing phase in which the system is preparing a response. When NAO speaks, the eyes turn white, then return to green at the end of its turn—signaling the start of a new interaction cycle.

Face tracking is also activated during phases in which NAO must show attentiveness—specifically, while listening to the user and while speaking to them—thus reinforcing the perception of engagement and responsiveness.

To ensure the interaction remains fluid and engaging, particular attention was paid to minimizing response times through concurrent processing. For example, with a user audio input of approximately 5 seconds, the system achieves around 3.5 seconds for audio upload and 2 seconds for parsing and response generation, resulting in a total system speaking response time of about 6 seconds. This is generally perceived as acceptable by users, especially considering that the perceived wait time is often shorter due to the continuous visual and behavioral feedback provided by NAO.

This extended framework demonstrates the feasibility of creating dynamic, emotionally expressive interactions that can run seamlessly on both the physical robot and its digital twin, paving the way for more complex comparative studies in social HRI. The architectural choices ensure that despite platform-specific implementation differences (e.g., microphone source, speech output method), the core logic—input, LLM processing, and the link between semantic content and emotional expression—remains consistent.

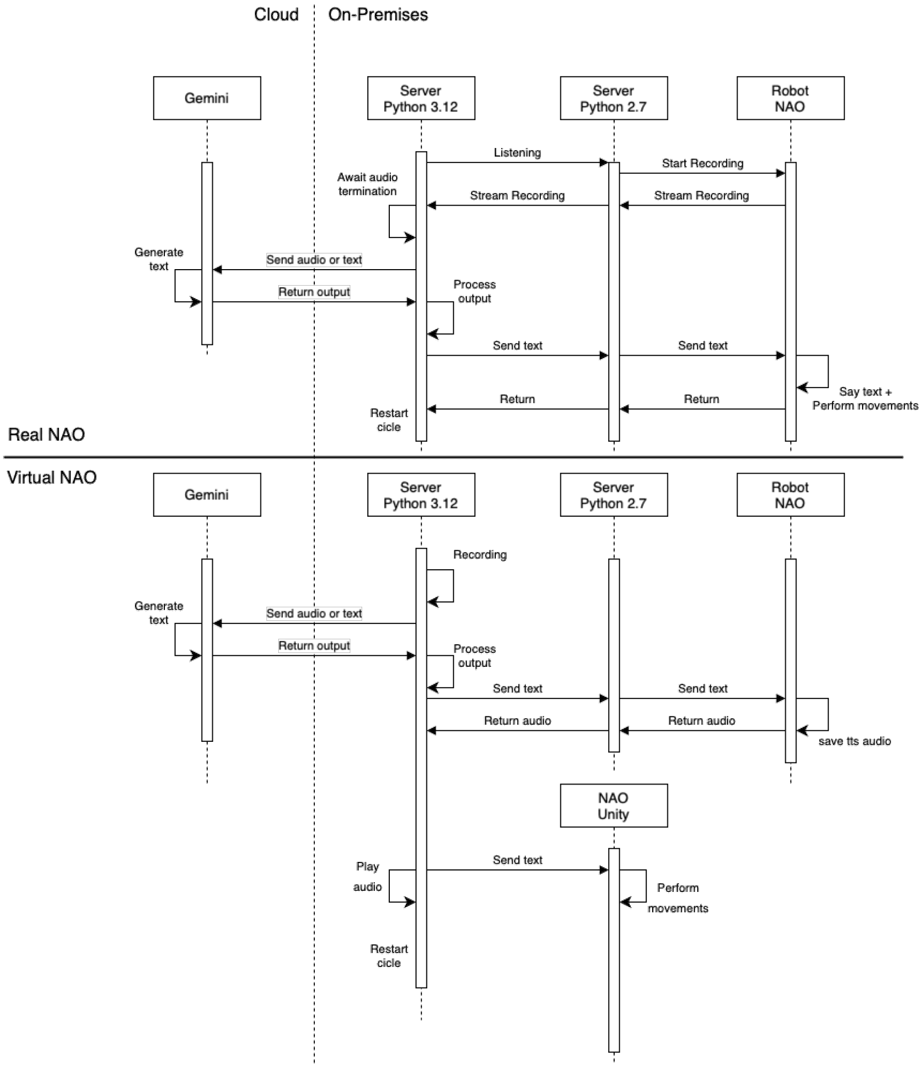

Figure 8: Sequence diagram of the extended framework including the LLM. User input is processed by Gemini to generate an empathetic response and an emotional cue, which are then enacted by the robot.

3.2 Experimental Procedure

This section details the user study conducted to compare human emotion recognition between a physical NAO robot and its high-fidelity digital twin. The experiment was structured as a within-subjects, counterbalanced comparison, where each participant interacted with both robot embodiments in two distinct, side-by-side stations. The procedure was designed to be robust and standardized, yet the stimulus presentation was randomized to ensure a natural and unbiased user experience.

3.2.1 Experimental Design and Stimuli

To investigate the impact of physical embodiment on emotion recognition, a within-subjects, counterbalanced experimental design was employed to mitigate order effects. Each participant interacted with both the physical and virtual NAO robot, experiencing all emotional stimuli.

Emotional Models and Selection

This study was grounded in Paul Ekman's model of basic emotions [7], which posits that certain emotions are universally recognized across cultures. Four emotions from Ekman's set were selected for this experiment—happiness, sadness, anger, and fear. These emotions are commonly used in human-robot interaction research and are within the expressive capabilities of the NAO robot, which relies on body gestures rather than facial expressions. Disgust was excluded because it is predominantly conveyed through facial expressions, which the NAO robot cannot reproduce effectively. Surprise was also omitted, as it is considered a complex emotion that can carry both positive and negative valence, complicating its interpretation in experimental settings [14, 15].

In addition to categorical emotion recognition, participants also rated the emotional stimuli along the dimensional axes of valence (pleasantness) and arousal (intensity) using the Self-Assessment Manikin (SAM) scales [26]. This allowed for a more nuanced understanding of emotional perception beyond simple categorization, aligning with Russell's circumplex model of affect [8], which maps emotions in a two-dimensional space. While these categorical and dimensional approaches may seem distinct, they are highly complementary. As illustrated in Figure 9, the discrete emotions of Ekman's model can be effectively plotted onto the valence-arousal space of the circumplex model, showing how, for example, happiness corresponds to high valence and high arousal, while sadness corresponds to low valence and low arousal.

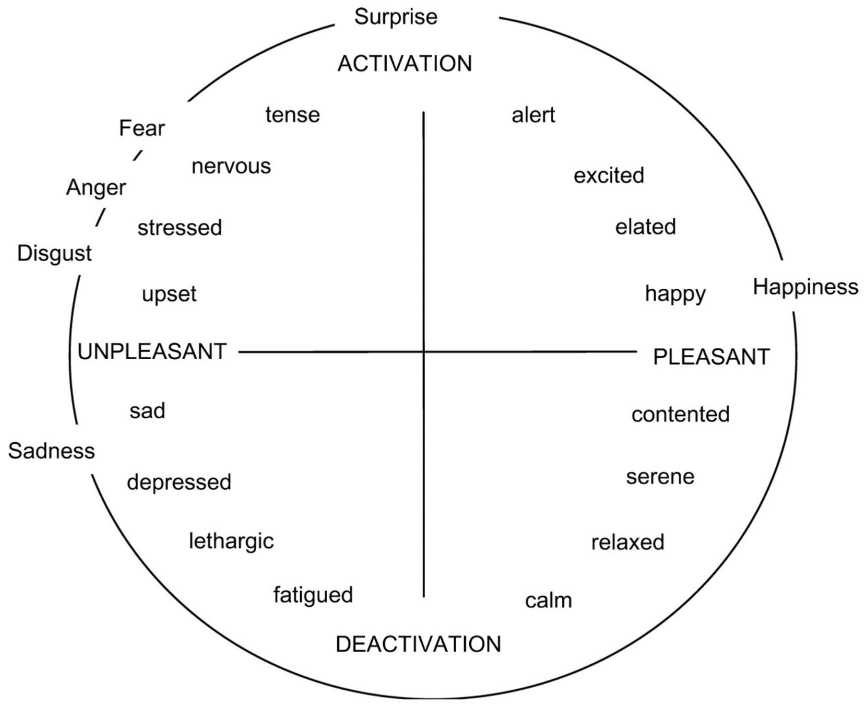

Figure 9: Russell's model with Ekman's six emotions—Happiness, Sadness, Anger, Fear, Surprise, Disgust.

Stimulus Set and Distribution

A total of 12 unique emotional gestures were implemented using SoftBank Robotics' Choregraphe software. Each emotion—happiness, sadness, anger, and fear—was represented by three distinct gestures, allowing for dynamic expressions through keyframe animations rather than static poses. This broader set of gestures ensured a more diverse and balanced distribution of stimuli, enabling pseudo-random presentation and minimizing response predictability.

The selection of the specific gestures was not arbitrary but was systematically grounded in the existing body of HRI literature, with a strong focus on studies involving the NAO robot. The primary goal was to select gestures that have been previously validated or frequently used in similar emotion recognition studies, thereby increasing the likelihood of their recognizability and providing a solid foundation for the experiment. The process involved a comprehensive literature review to identify validated postures and dynamic gestures for each of the four target emotions. Table 1 summarizes this selection process, cross-referencing each of the 12 chosen gestures with the studies in which they were employed.

Once selected, each of these 12 gestures was meticulously implemented in the Choregraphe software. The resulting set of animations, as rendered on the high-fidelity digital twin, is presented in Figure 5. To visually confirm the behavioral parity—a core tenet of this study—Figure 10 shows the physical NAO robot executing the exact same set of gestures. This side-by-side comparison highlights the successful replication of the emotional expressions across both embodiments.

To ensure a comprehensive and non-repetitive experience, each participant was exposed to all 12 gestures exactly once throughout the entire experiment. The 12 gestures were randomly divided into two sets of six for each participant's session. The order of conditions (Physical vs. Virtual) and the assignment of gesture sets to each condition were counterbalanced across participants. This ensured that while the experiment felt entirely random to the user, the overall design remained balanced.

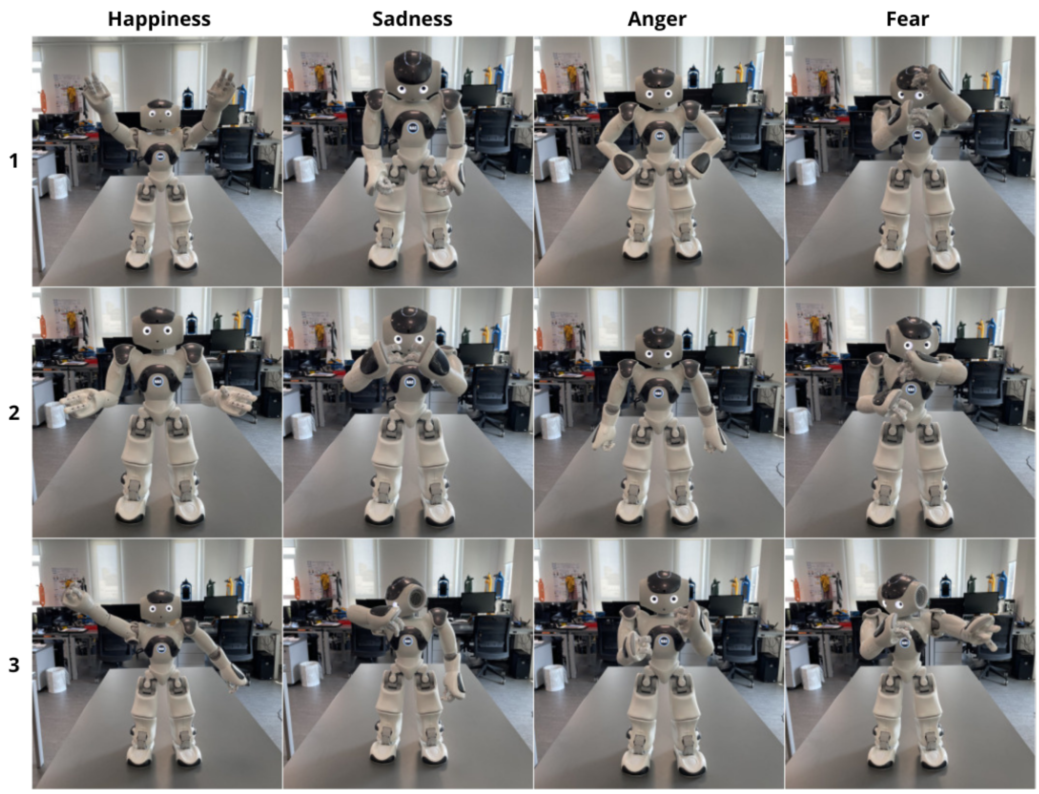

Figure 10: Emotion gestures: Happiness, Sadness, Anger, Fear. Three gestures for each emotion. It allows for a direct comparison with the virtual robot's gestures shown in Figure 5.

3.2.2 Apparatus and Environment

The experiment was conducted using two side-by-side stations set up in a quiet room to minimize distractions.

- **The Physical Station** consisted of a SoftBank Robotics NAO V6 robot placed on a table.
- **The Virtual Station**, positioned approximately one meter away, featured a 24-inch monitor displaying the high-fidelity digital twin rendered in the Unity engine.
- **Control and Interface Hardware:** A chair was placed in front of each station, and participants were provided with a tablet to access the data collection interface. Participants were free to adjust their position but were instructed to maintain a consistent viewing perspective for both conditions to ensure a fair comparison. The core server logic was hosted on a MacBook Pro (M1, 16GB RAM), which was connected to the same local network as the NAO robot and the participant's tablet. The monitor for the virtual station was connected to this laptop via HDMI.

3.2.3 Measures and Data Collection

Data was collected through a custom web-based user interface displayed on the tablet (see Figure 11).

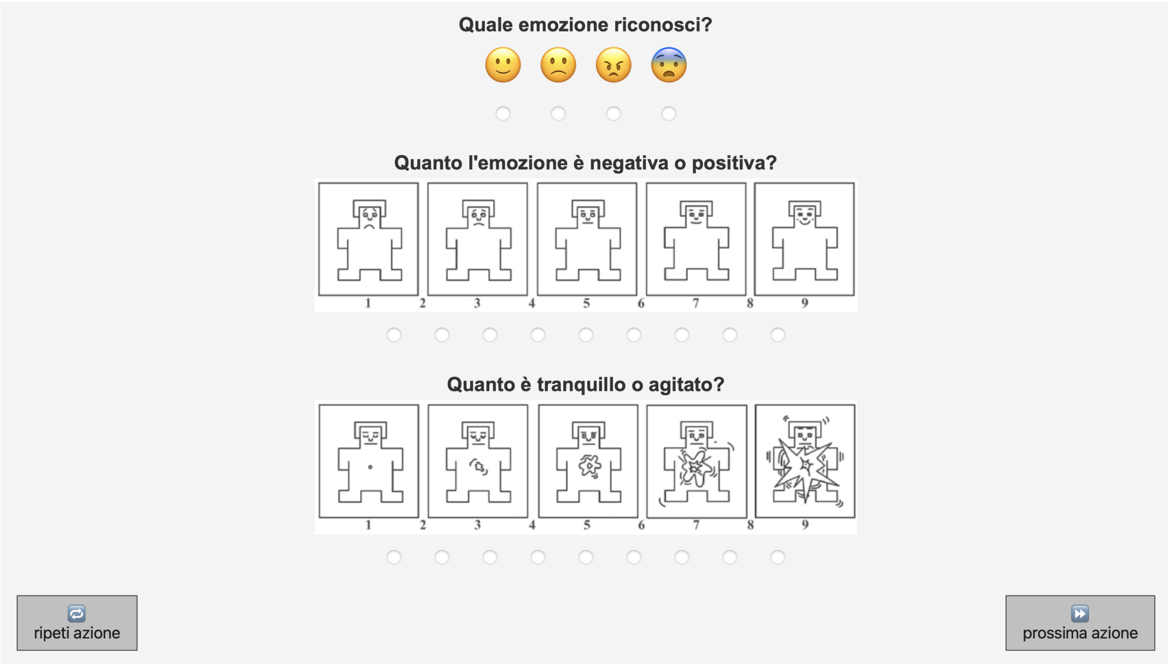

Figure 11: Screenshot of the user interface for the data collection task, showing the forced-choice emotion classification question and the Self-Assessment Manikin (SAM) scales for valence and arousal.

- **Emotion Classification Task:** After each gesture, participants were presented with a forced-choice question. Prior to starting, participants were explicitly informed about which emotion corresponded to each of the four emojis (in order: Happiness, Sadness, Anger and Fear).

- **Self-Assessment Manikin (SAM):** The SAM was used to capture the dimensional aspects of the perceived emotion. Unlike a simple classification, SAM provides quantitative data on two fundamental axes of emotion: **Valence** (the degree of pleasantness, from negative to positive) and **Arousal** (the level of intensity, from calm to excited). This methodology was chosen to provide a richer, more nuanced understanding of the participant's perception beyond a simple categorical label [26].

- **Data Logging:** All participant responses were automatically recorded and stored in a structured JSON file. To ensure anonymity, each experimental session assigned a randomly generated universally unique identifier (UUID) to the participant's data set. This UUID served as the root key in the JSON file, completely decoupling the collected data from any personal identifiers. Figure 12 shows the structure of one such data file.

```
{
    "84eb6475-43bb-4506-9ef9-1ca722e69650": [
        {
            "interaction": "virtual",
            "emotion": "Fear1",
            "emotion-recognized": "fear",
            "valence": "3",
            "arousal": "6"
        },
        // ... (5 more virtual interactions)
        {
            "interaction": "real",
            "emotion": "Happiness1",
            "emotion-recognized": "happy",
            "valence": "8",
            "arousal": "7"
        }
        // ... (5 more real interactions)
    ]
}
```


Figure 12: Example structure of an anonymized data file. A list of all 12 interactions, detailing the condition ('interaction'), the specific gesture shown ('emotion'), the participant's classification ('emotion-recognized'), and their SAM ratings. This structured output facilitates subsequent statistical analysis.

3.2.4 Step-by-Step Protocol

The experimental procedure was identical for all participants, apart from the counterbalanced condition order. The entire session lasted approximately 10-15 minutes.

1. **Introduction to the study:** Participants were welcomed and introduced to the general purpose of the study, without revealing the specific hypotheses. They were then handed a tablet and instructed on how to use the interface.

2. **First Condition:** The experiment began with the first condition, either physical or virtual. The robot then performed the first of six pseudo-randomly selected emotional gestures from its assigned set.

3. **Data Collection per Trial:** Following each gesture, the interface on the tablet prompted the participant for a response.

- At any point during a trial, before proceeding to the next gesture, participants could re-watch the current animation as many times as they wished by pressing the “repeat action” button.
- Once ready, they classified the observed emotion by selecting one of the four emoji options.
- They then rated the emotion’s valence and arousal using the respective SAM scales.
- After submitting their ratings by pressing the “next action” button, the system would automatically trigger the next trial.

4. **Completing the Condition:** This trial process was repeated for all six unique gestures assigned to the first condition.

5. **Second Condition:** The procedure was then repeated identically for the second condition, presenting the remaining six unique emotional gestures performed by the other robot embodiment. After the final trial, the experiment concluded.

6. **Briefing:** At the end of the experiment, participants who wished to share their thoughts were invited to provide personal feedback. Finally, the full purpose of the study was explained to them.

Figure 13: Comparison between the physical (left) and virtual (right) robot setups. Both environments were carefully matched in appearance and configuration to ensure high-fidelity parity for experimental control.

Chapter 4

Results

This chapter presents the results of the experimental study conducted to compare human emotion recognition between a physical NAO robot and its digital twin. The data were collected from 24 participants and analyzed to assess recognition accuracy and the dimensional perceptions of valence and arousal.

4.1 Participants

A total of 24 individuals participated in the study. All participants were neurotypical adults, aged between 18 and 35, and were recruited from the university community. Participation was voluntary, and all collected data were fully anonymized to protect participant privacy.

To mitigate order effects, a counterbalanced within-subjects design was employed. Participants were pseudo-randomly assigned to one of two starting conditions:

- 13 participants (54.2%) began the experiment with the physical robot (*real* condition).
- 11 participants (45.8%) began the experiment with the virtual robot (*virtual* condition).

Each participant evaluated a total of 12 unique emotional gestures—6 in the *real* condition and 6 in the *virtual* condition—for a total of 288 collected observations (24 participants $\times$ 12 observations). Full data are available in the Appendix 7.2.

4.2 Emotion Recognition Accuracy

The primary analysis focused on the accuracy with which participants identified the emotion expressed by the robot.

4.2.1 Overall Accuracy and Confusion Matrix

Considering all 288 observations, the overall mean recognition accuracy was **76.7%**. Figure 14 shows the general confusion matrix, which illustrates how emotions were classified. The rows represent the true emotion (stimulus), and the columns represent the emotion perceived by the participant.

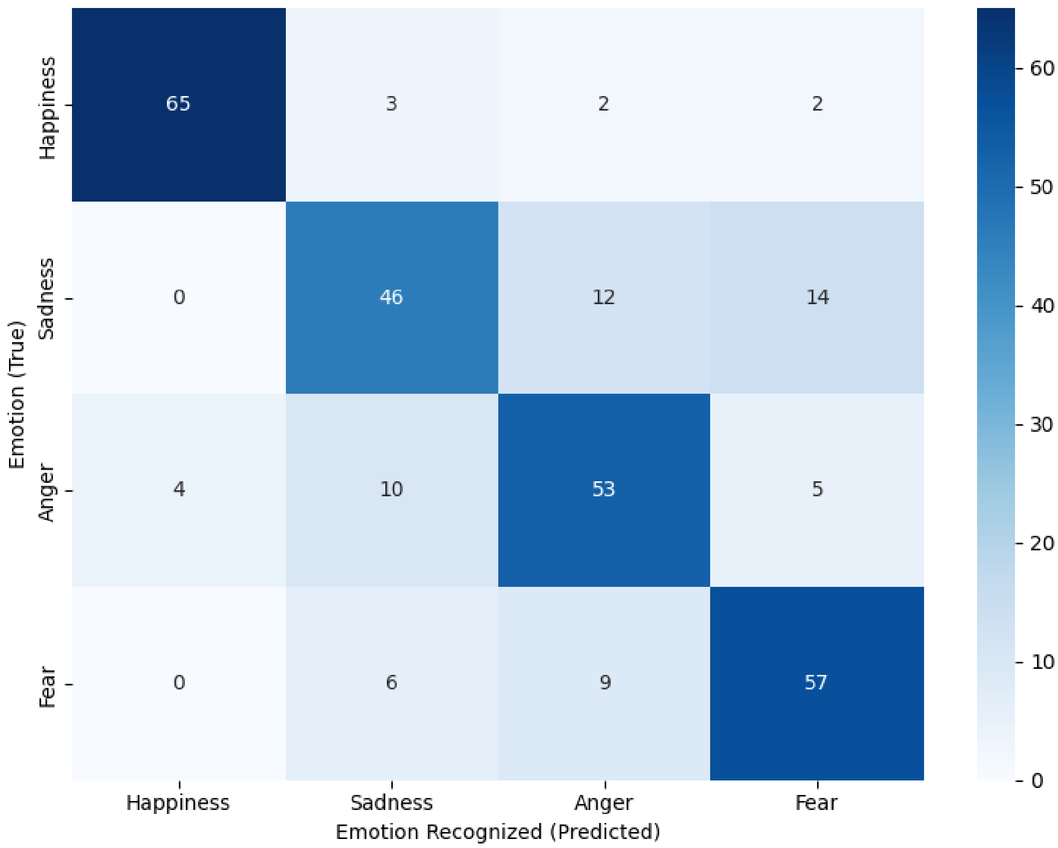

Figure 14: Overall confusion matrix.

From the general confusion matrix, it is observed that confusions occur almost exclusively between emotions of the same valence. *Happiness* is rarely misidentified, whereas the negative-valence emotions are frequently mistaken for one another. For instance, *Sadness* is often confused with *Fear* (19.4% of instances) and *Anger* (16.7%).

4.2.2 Accuracy by Condition: Physical vs. Virtual

The study's central hypothesis was that the robot's physical embodiment would influence emotion recognition accuracy. To test this, mean accuracy was calculated for each of the two experimental conditions:

- Accuracy in the **Real** Condition: **77.8%**
- Accuracy in the **Virtual** Condition: **75.7%**

To determine if the observed 2.1% difference in accuracy was statistically significant or simply a result of random variation, a paired-samples t-test was conducted. This is the appropriate analysis for a within-subjects design, as it directly compares the performance of each participant across the two conditions. The test assesses whether the difference between the real and virtual conditions is large enough to be considered reliable, given the variability in participant responses. It is possible to do this test since the differences between the two conditions follow a Gaussian distribution.

The Paired-Samples T-Test: A paired-samples t-test was conducted to rigorously assess whether the difference in accuracy between the real and virtual robot conditions was statistically significant. The results showed no significant effect: $t(23) = 0.44$, $p = .66$.

- **The t-statistic (t):** The t-statistic measures the size of the difference between two related conditions, relative to the variability in the data. In this study, the t-value reflects the size of the mean difference in accuracy between the 'Real' and 'Virtual' conditions relative to the variability of that difference among participants. A high t-value would indicate a clear and consistent difference, whereas a value close to zero, such as the one obtained (t = 0.44), suggests that the observed difference is small compared to random fluctuations in individual performance.
- **The p-value (p):** The statistical test begins with the null hypothesis—that there is no difference in performance between the physical and virtual robot conditions. The p-value represents the probability of obtaining the observed difference (or a more extreme one) if this assumption were true. In our case, a p-value of .66 indicates that, assuming no real difference exists between the two conditions, there is a 66% probability of obtaining a difference as large or larger than the one observed, purely by chance. Since this probability is high—well above the conventional threshold of 5% ($p < .05$)—we cannot reject the null hypothesis. Therefore, we do not have statistical evidence indicating that one condition is superior to the other.

This outcome supports the hypothesis that both platforms function equivalently in this task, reinforcing the validity of the virtual robot as a reliable substitute for its physical counterpart.

The confusion matrices for each condition, presented in Figure 15, show similar patterns of error, reinforcing the conclusion that embodiment did not systematically alter how emotions were perceived or misperceived.

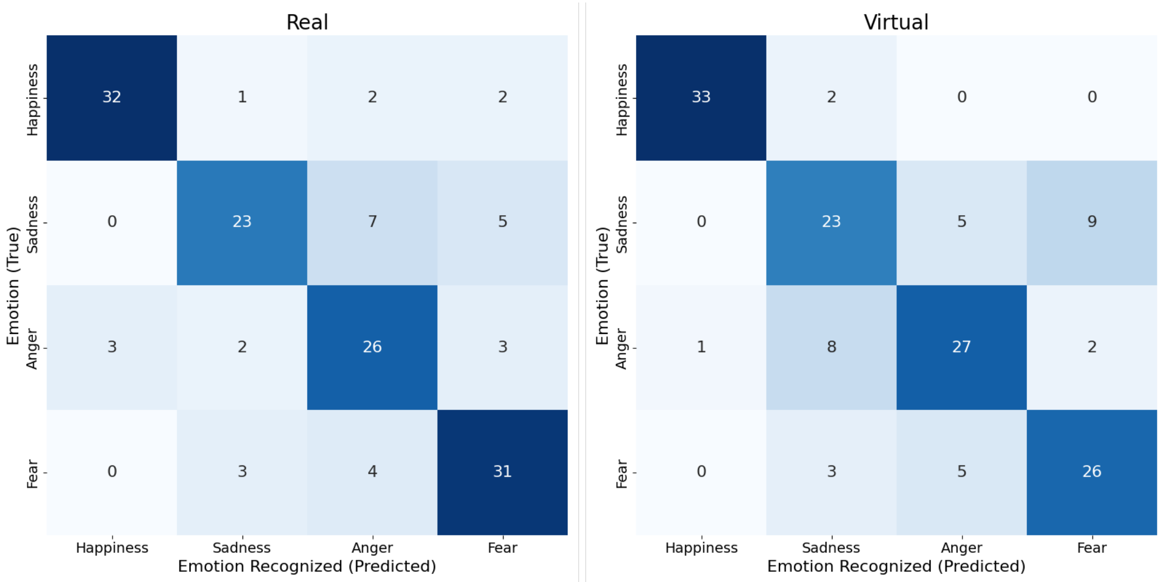

Figure 15: Real vs Virtual confusion matrices.

4.2.3 Accuracy by Emotion Category and Individual Stimulus

The recognition accuracy varied substantially across the four emotion categories. As shown in Table 2, *Happiness* was the most recognized emotion (90.3%), followed by *Fear* (79.2%), *Anger* (73.6%), and finally *Sadness* (63.9%).

Stimulus	Acc. Real (%)	Errors	Acc. Virtual (%)	Errors	Acc. Combined (%)
Happiness1	100.0	–	100.0	–	100.0
Happiness3	100.0	–	100.0	–	100.0
Happiness2	61.5	angry (2), fear (2), sad (1)	**81.8**	sad (2)	70.8
Fear1	**100.0**	–	90.9	angry (1)	95.8
Fear2	76.9	angry (2), sad (1)	**90.9**	sad (1)	83.3
Fear3	**66.7**	sad (2), angry (2)	50.0	angry (4), sad (2)	58.3
Anger3	84.6	fear (2)	**90.9**	fear (1)	87.5
Anger1	66.7	happy (2), sad (1)	**93.3**	sad (1)	83.3
Anger2	**75.0**	happy (1), fear (1), sad (1)	25.0	sad (7), happy (1), fear (1)	50.0
Sadness1	**80.0**	angry (2)	78.6	fear (2), angry (1)	79.2
Sadness3	**75.0**	fear (2), angry (1)	58.3	fear (4), angry (1)	66.7
Sadness2	**46.2**	angry (4), fear (3)	45.5	fear (3), angry (3)	45.8

Table 2: Recognition accuracy for each stimulus. The bold value indicates the condition with higher accuracy. Alongside accuracy, the associated errors that led to mistakes.

However, a more granular analysis reveals that communicative efficacy depends more on the specific gesture itself rather than on the broader category. Table 2 details the performance of each of the 12 unique stimuli, highlighting a significant variability. This analysis is crucial as it shifts the focus from the embodiment to the quality of the interaction design itself. Gestures with strong, unambiguous semantics (e.g., 'Happiness1', 'Fear1') achieved near-perfect accuracy, whereas ambiguous gestures (e.g., 'Sadness2') performed poorly, whether it was seen in the physical or virtual robot.

4.2.4 In-depth Analysis of Embodiment and Emotion Type

To move beyond a simple comparison and investigate the interplay between the robot's embodiment and the type of emotion being expressed, a 2 (Condition: Real, Virtual) x 4 (Emotion Category: Happiness, Sadness, Anger, Fear) repeated measures Analysis of Variance (ANOVA) was conducted on the recognition accuracy data. This analysis allows for a more nuanced understanding of how these two factors influence user perception. Also in this case we can use this test since the differences between real and virtual follow a Gaussian.

The analysis yielded three key findings:

1. **Effect of Condition:** On average, participants were equally adept at identifying emotions regardless of the robot's embodiment. The ANOVA revealed no significant main effect of Condition, $F(1, 23) = 0.05, p = .82$. This result confirms the paired-samples t-test, providing robust evidence that overall recognition accuracy did not differ between the physical and virtual robot conditions. The slight difference in test statistics ($p = .66$ for the t-test) is due to the ANOVA's more nuanced calculation, which accounts for variance from other factors.

2. **Effect of Emotion Category:** A statistically significant main effect of Emotion

Category was found, $F(3, 69) = 3.74, p = .015$. This indicates that the likelihood of an emotion being correctly identified depended heavily on which emotion was being expressed. This finding highlights that the intrinsic clarity and design of the emotional gestures were a more dominant factor in recognition success than the robot's physical form.

3. **Effect of Condition x Emotion Category:** Crucially, the interaction effect between Condition and Emotion Category was not significant, $F(3, 69) = 0.12, p = .95$. This is a powerful null result, as it indicates that the (non-existent) difference between the real and virtual conditions was consistent across all four emotion categories. In other words, there is no statistical evidence to suggest that the robot's physical embodiment provided a specific advantage for recognizing any single emotion, such as a more ambiguous or complex one, over others.

In summary, the ANOVA results reinforce the conclusion that the physical presence of the robot did not influence emotion recognition. Instead, the primary driver of recognition success was the inherent communicative quality of the specific gestures used for each emotion.

4.3 Dimensional Analysis: Valence and Arousal (SAM)

In addition to the categorical classification, participants rated each emotion on the dimensional axes of Valence (1=very unpleasant to 9=very pleasant) and Arousal (1=very calm to 9=very excited) using the Self-Assessment Manikin (SAM) questionnaire. This analysis provides insight into the qualitative perception of the emotions.

True Emotion	**Mean Valence (SD)**	**Mean Arousal (SD)**
Happiness	7.28 (1.52)	4.42 (2.38)
Sadness	2.69 (1.39)	5.00 (2.01)
Anger	3.43 (1.69)	5.06 (2.05)
Fear	2.74 (1.45)	6.67 (1.92)

Table 3: Mean Valence and Arousal scores (and standard deviation) for each emotion category, aggregated across conditions.

The dimensional scores (Table 3) are consistent with Russell's circumplex model, with *Happiness* high in valence and the other three low. Notably, *Fear* is perceived as the highest-arousal emotion, while *Sadness* and *Anger* occupy an intermediate arousal space. This helps explain the confusion patterns: participants correctly identify the negative valence of ‘Sadness‘, ‘Anger‘, and ‘Fear‘, but struggle to differentiate them based on their perceived level of intensity or arousal.

Figure 16 displays the 12 individual stimuli in Valence–Arousal space, visually confirming both the clustering of emotions and the ambiguity of certain gestures. As can be observed, the more accurately an emotion was recognised (shown by a darker shade within its category’s colour), the closer it is plotted to its theoretically correct position on Russell’s circumplex. Conversely, when participants misidentified an emotion, they tended to place their choice farther from the expected location.

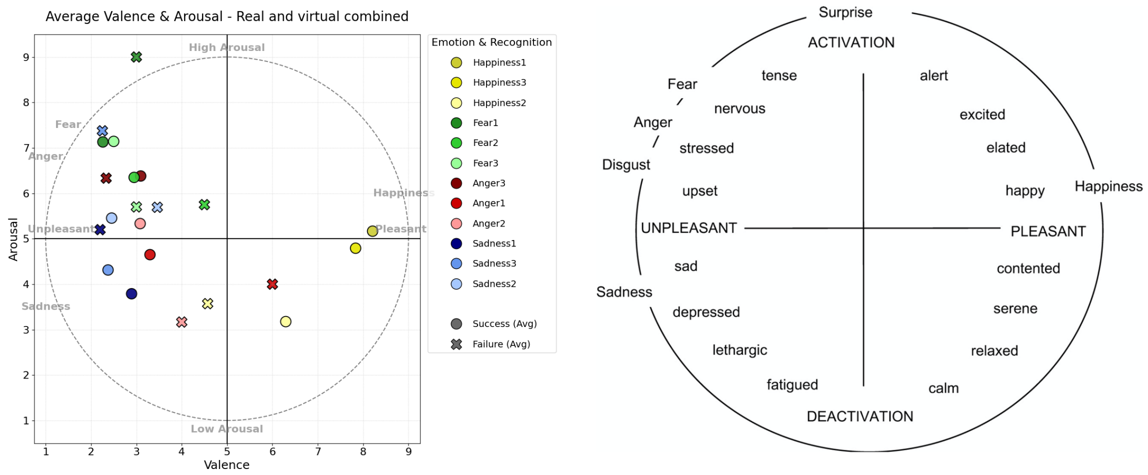

Figure 16: Valence–arousal maps for emotion-recognition outcomes for both conditions. Dots represent the average coordinates of the correctly recognized emotions, the cross of those wrong for that same emotion. The dashed circle illustrates Russell’s circumplex, while thick lines divide the space into the four canonical quadrants. The side-by-side image represents the original Russell’s circumplex.

4.3.1 Emotion Distribution by Condition and Recognition Accuracy

When we contrast the correct and incorrect recognition panels for both the Real and Virtual conditions, a clear pattern emerges. Correct responses cluster tightly around the sector of Russell’s circumplex that theoretically corresponds to each target emotion. By contrast, incorrect responses do not drift randomly across the map; instead, they gather near the sector associated with the label actually selected by observers. For instance, the stimulus “Anger1” was misclassified as happiness twice in the real condition, its dot appears near the happiness region. This pattern is especially evident for stimuli that generate many errors: those dots tend to sit squarely in the region corresponding to the “wrong” label, signaling that the underlying gesture conveys affect that is genuinely hard to differentiate. Overall, the slight increase in dispersion reflects uncertainty rather than random guessing, and it confirms that embodiment (physical versus virtual) does not systematically distort where people locate the robot’s expressions in the circumplex.

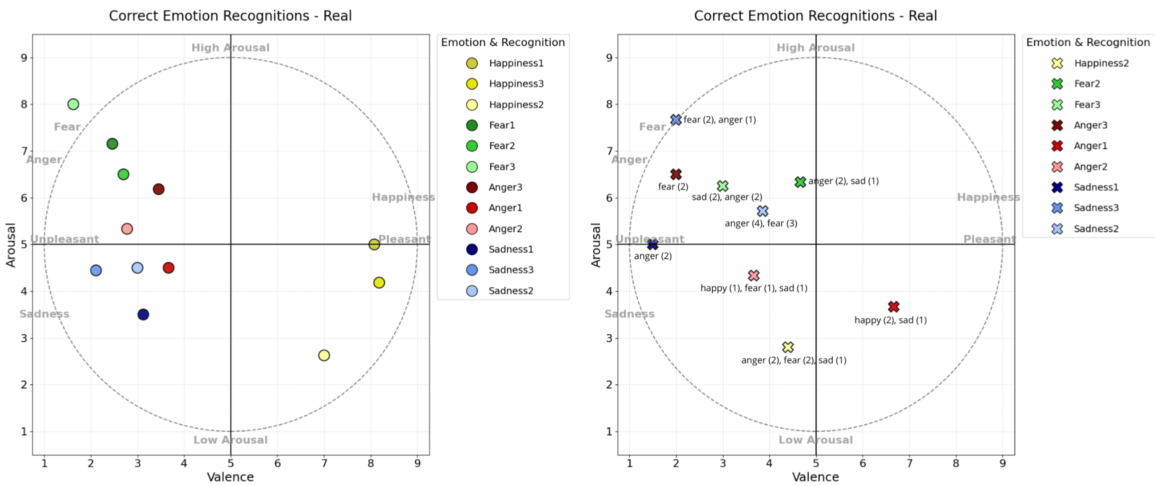

Figure 17: **Real condition** - The left panel displays cases where the emotions were correctly recognized, while the right panel shows instances where the emotions were misrecognized. Text labels indicate the actual emotions recognized, with the number of occurrences shown in parentheses (e.g., ‘anger (2)’ means the stimulus was incorrectly classified as anger twice).

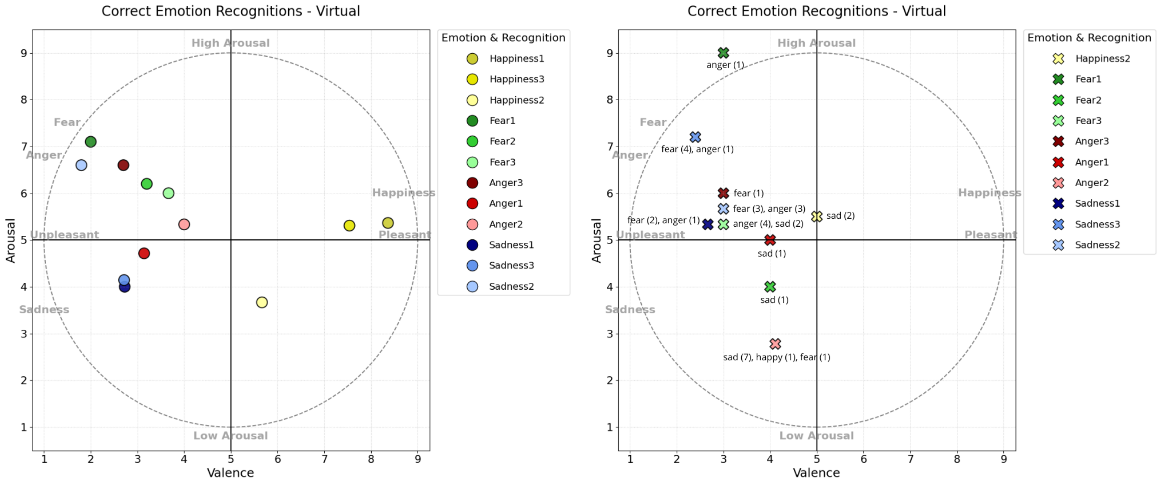

Figure 18: **Virtual condition** - The left panel displays cases where the emotions were correctly recognized, while the right panel shows instances where the emotions were misrecognized. Text labels indicate the actual emotions recognized, with the number of occurrences shown in parentheses (e.g., 'anger (2)' means the stimulus was incorrectly classified as anger twice).

4.4 Qualitative User Feedback

Beyond the quantitative data, qualitative insights were gathered through informal briefing with participants following the experiment. These unstructured discussions provided an opportunity for participants to share open feedbacks about their experience with both the real and virtual robot conditions. While this feedback was not structured around specific prompts, it provides valuable insights into the subjective user experience.

A notable theme emerged concerning participant comfort. Five participants explicitly commented that they found the interaction with the real robot to be more "comfortable" than the interaction with its virtual counterpart, reporting to prefer interaction with a physical agent instead of a digital one.

It is important to contextualize this finding. This feedback was unsolicited, and no formal, validated scale for comfort or user experience was administered. Therefore, this observation should be considered exploratory and indicative of a potential experiential difference, rather than a statistically robust conclusion. It suggests future validation using standardized instruments to assess user interaction and perceived comfort (This was done in the preliminary study involving the integration of LLM into the digital/real embodiment: see Section 4.5).

4.5 Preliminary NAO LLM Assessment

In addition to the main experiment, a preliminary, exploratory study was conducted to evaluate the LLM-integrated dialogue framework described in Section 3.1.5. The primary goal of this small-scale assessment was not to produce statistically significant results, but to serve as a proof-of-concept, test the real-world feasibility of the system, and gather initial user impressions regarding the interaction with an empathetic, conversational robot in both its physical and virtual forms.

4.5.1 Procedure

A total of four participants were recruited for this pilot study. The participants were divided into two groups: Two participants engaged in an open-ended dialogue with the physical NAO robot, the other two with the virtual counterpart.

The interaction was designed to be natural and spontaneous. To facilitate conversation and provide some direction, participants were given a sheet 7.3 with suggested topics and questions they could use to initiate or guide the dialogue with the robot. This strategy was particularly important to avoid stilted, question-answer exchanges in which participants—especially at the beginning when they might feel awkward—tend to ask very closed, factual questions such as "What's your name?", "Who are you?", or "Do you like reading?". These types of questions limit the development of a meaningful dialogue and constrain the robot's ability to express affective or social behavior. By providing broader and more engaging prompts, the design aimed to foster a more open conversation where the robot could display a range of social attributes and emotional expressions, making the interaction more immersive and human-like. Following the interaction, each participant was asked to complete the Robotic Social Attributes Scale (ROSAS) questionnaire to evaluate their perception of the robot. The questionnaire used is visible in the Appendix 7.4.

4.5.2 Measure: The Robotic Social Attributes Scale (ROSAS)

To measure participants' perceptions of the robot, the Robotic Social Attributes Scale (ROSAS) was used [27]. ROSAS is a validated instrument designed to assess a user's impression of a robot's social characteristics. Participants rate the robot on a series of adjectives using a 5-point Likert scale (from 1 = "Not at all" to 5 = "Very much"). The scale is structured around three core subscales:

- **Competence:** Assesses the robot's perceived capability, reliability, and intelligence ("Competent", "Capable", "Responsible", "Interactive", "Reliable", "Knowledgeable").

- **Warmth:** Evaluates the robot's perceived social and emotional qualities ("Happy", "Feeling", "Social", "Compassionate", "Emotional", "Organic").
- **Discomfort:** Measures negative feelings or perceptions elicited by the robot ("Scary", "Strange", "Awkward", "Dangerous", "Awful", "Aggressive").

4.5.3 Preliminary Findings

To provide a quantitative overview of the ROSAS results, the mean scores for each of the 18 items were calculated for both the real and virtual conditions. Figure 19 visualizes these averages, grouping the items by their respective subscales: Competence, Warmth, and Discomfort.

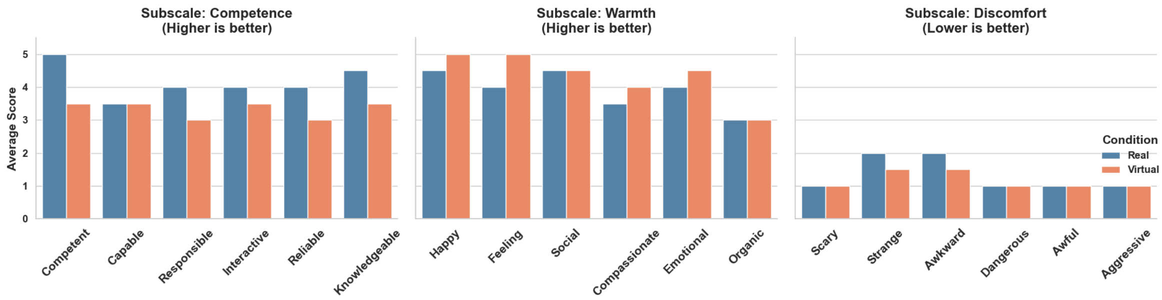

Figure 19: Comparison of mean Robotic Social Attributes Scale (ROSAS) scores for the Real and Virtual robots. The bars represent the average rating for each item on a 5-point scale, grouped by the three core subscales.

As illustrated in the figure, a consistent trend emerges from this preliminary data. For the **Competence** subscale, the physical robot was generally rated higher than its virtual counterpart across most items. Oppositely, for **Warmth** subscale the virtual counterpart had better scores. For the **Discomfort** subscale, both conditions received very low scores, indicating that neither robot was perceived as strange, awkward, or threatening by the participants. However, these patterns are not statistically significant due to the small sample size, future studies will need to meticulously validate this trend.

Qualitative Comments

In addition to the scaled ratings, two participants also tried the other version of NAO and provided an open written feedback, which offers valuable context for the quantitative scores.

- **Participant who started with real condition:** Wrote, *"The real interaction is more pleasant, more engaging, it adds a lot to the interaction"*.

- **Participant who started with virtual condition:** Wrote, *"I preferred to dialogue with the real NAO, it seemed like I was talking to a 'more real person' compared to the virtual NAO, where it gave me more of the impression of talking to a machine. Also, the fact of having a gaze that follows you is perceived more with the real NAO; with the virtual NAO I was more easily distracted, it was less engaging".*

These patterns, while not statistically significant due to the small sample size, align with the qualitative feedback collected for the main experiment and suggest that physical embodiment may positively influence perceptions of a robot's social and functional capabilities in a conversational context.

Chapter 5

Discussion

This chapter provides an in-depth interpretation of the experimental results presented in Chapter 4. The findings are discussed in the context of the initial research questions and the existing literature on Human-Robot Interaction (HRI) and embodiment. The primary aim of this study was to empirically investigate whether the physical embodiment of the NAO robot influences the accuracy and perception of its emotional expressions compared to a visually identical, high-fidelity digital twin (see Section 2.4). This discussion will analyze the key findings, address the core hypotheses, and explore the theoretical and practical implications of the results.

5.1 The Null Effect of Embodiment on Recognition Accuracy

The core hypothesis of this thesis was that the physical presence of the NAO robot would enhance the accuracy of emotion recognition compared to its virtual twin. The results failed to support this hypothesis. The repeated-measures ANOVA 4.2.4 revealed no significant main effect between the two conditions ($F(1, 23) = 0.05,\ p = .82$), a finding reinforced by the initial paired-samples t-test. The marginal 2.1% difference in mean accuracy (77.8% for Real vs. 75.7% for Virtual) was statistically indistinguishable from random chance.

This null result is the most direct answer to our primary research question and has significant implications. As outlined in our motivation, a key goal was to determine if less expensive and more scalable virtual simulations could serve as valid proxies for physical robot experiments in the affective domain. Our findings strongly suggest that, for the specific task of categorical emotion recognition based on bodily gestures, a high-fidelity virtual twin is indeed a valid substitute. This represents a meaningful contribution to the state of the art, suggesting that for specific tasks such as emotion recognition, this type of substitution is indeed feasible. This aligns with recent research, such as the meta-analysis by Esterwood et al. [4], which found no significant differences in some specific tasks between physically collocated and non-collocated robots. Our work provides focused, empirical evidence to support this view specifically within the domain of emotional expression.

5.2 The Dominant Role of Emotion Category

While embodiment did not prove to be a significant factor, the type of emotion being expressed was a powerful determinant of success. The ANOVA revealed a statistically significant main effect of Emotion Category ($F(3, 69) = 3.74,\ p = .015$). This indicates that the intrinsic recognizability of the gestures varied dramatically from one emotion to another.

As detailed in Table 2 and the confusion matrices in Figure 15, Happiness was the most easily recognized emotion (90.3%), whereas the negative-valence emotions—Anger (73.6%), Fear (79.2%), and especially Sadness (63.9%)—were frequently confused with one another. This pattern is consistent with established literature on emotional expression. Gestures for happiness are often large, distinct, and culturally unambiguous. In contrast, bodily expressions of fear, sadness, and anger can share common features (e.g., slumped posture, contracted movements), making them inherently more ambiguous, particularly in the absence of nuanced facial expressions, which the NAO robot lacks.

Crucially, this finding shifts the focus of the embodiment debate. The critical factor for successful emotional communication in this context was not the robot's physical or virtual nature, but the inherent clarity and quality of the gesture design itself. The significant variability in accuracy for stimuli within the same category (e.g., Fear3 at 58.3% vs. Fear1 at 95.8%) underscores this point. Unambiguous, well-designed gestures (Happiness1, Fear1) achieved near-perfect recognition in both conditions, while ambiguous ones (Sadness2) performed poorly in both.

This conclusion is powerfully reinforced by the lack of a significant interaction effect between Condition and Emotion Category ($F(3, 69) = 0.12,\ p = .95$). It demonstrates that the physical robot did not offer a special advantage for recognizing any particular emotion, not even for those that proved more ambiguous or difficult to classify. The (non-existent) performance gap between the real and virtual conditions remained consistent across all four emotion categories, reinforcing the conclusion that embodiment, in this context, was not a critical variable.

5.3 Interpreting Confusion Patterns through a Dimensional Analysis

The overall confusion matrix in Figure 14 showed that errors were not random; they occurred almost exclusively between emotions of the same negative valence. Sadness, for instance, was frequently confused with Fear and Anger.

The dimensional analysis of Valence and Arousal in Figure 16 provides a compelling explanation for this pattern.

As shown in Table 3, participants correctly identified the low valence of Sadness (2.69), Anger (3.43), and Fear (2.74) as opposed to Happiness(7.28). They knew the robot was expressing a "negative" emotion. However, they struggled to differentiate these emotions based on their perceived level of activation (Arousal), explaining the high degree of confusion between them.

This interpretation is further reinforced by examining the spatial distribution of the stimuli in the Valence-Arousal plots (Figures 17 and 18). When an emotion was correctly recognized, its corresponding point clustered tightly within the correct quadrant of Russell's circumplex. Crucially, when an emotion was misidentified, its point did not land randomly on the map; instead for most cases, it migrated squarely into the quadrant of the "wrong" label chosen by the participant. For example, when Sadness3 was mistaken for Fear, its rated valence and arousal placed it in the "high-arousal, negative-valence" quadrant, right where Fear belongs.

This demonstrates two critical points. First, it confirms that both the physical and virtual robots were successfully conveying affective information that participants could map onto Russell's model. Second, it shifts the blame for errors away from the participant's perception or the robot's embodiment, and places it squarely on the ambiguity of the gestures themselves. The problem was not that the robot failed to express an emotion, but that the gesture, for example, Sadness conveyed a level of arousal that was too similar to that of Fear.

5.4 Functional and Experiential Difference

A fascinating nuance emerged when contrasting the quantitative results with the qualitative user feedback 4.4. While the data clearly showed functional equivalence in task performance, five participants spontaneously reported feeling "more comfortable" interacting with the physical robot.

This highlights a critical distinction between functional equivalence and experiential equivalence. Functionally, for the task of identifying emotions, the virtual robot was a viable substitute. However, experimentally, the physical robot seemed to foster a different, more positive subjective state. This can be interpreted through the lens of social presence and proxemics. The tangible, co-located nature of the physical robot likely engendered a stronger sense of “being with” another entity, reducing the artificiality of the interaction and creating a more natural rapport. This provides a nuanced interpretation of earlier findings, such as those by Li et al. [2], which noted that virtual robots could be more “discomforting”. Our results suggest this perception is not tied to functional performance but to the subjective quality of the social experience itself.

This finding does not contradict the accuracy results; it enriches them. It implies that the choice of platform, while potentially irrelevant for measuring task performance, could be highly relevant for studies investigating long-term engagement, trust and social presence. For a researcher testing the legibility of a gesture, the virtual twin is sufficient. For creating a companion robot for the elderly, the subtle yet meaningful experiential benefits of physical presence, and the “comfort” it engenders, cannot be ignored.

5.5 Implications for LLM-Driven Empathetic Interaction

The findings from both the primary experiment and the preliminary pilot study have direct implications for this thesis's secondary objective: the creation of an effective, LLM-driven empathetic interaction framework. The development of such systems hinges on the seamless integration of emotional behaviours with conversational intelligence. To achieve this, our framework was designed to insert emotional tags into the LLM's output, enabling the robot—both physical and virtual—to exhibit non-verbal cues synchronized with its speech (see Section 3.1.5).

A key takeaway from our main experiment is the validation of the digital twin as a reliable proxy for the physical robot in the specific task of emotion recognition. This is a crucial practical advantage, as it allows for the rapid, scalable prototyping of LLM-generated emotional expressions in a more flexible virtual environment. We can confidently design and iterate on social cues with the digital twin, knowing that their perceptual clarity will effectively transfer to the physical robot.

The preliminary ROSAS assessment (Section 4.5) provides a first glimpse into how embodiment affects perceptions in this dynamic, conversational context. A fascinating and complex picture emerges from the data. The physical robot was perceived as more competent, despite being driven by the exact same LLM and control logic as the virtual one. This difference cannot be attributed to functional performance. Instead, it points to a powerful psychological effect of embodiment: the physical presence of the agent appears to lend it an aura of capability and reliability, aligning with the qualitative feedback where users described it as more ”real” and ”engaging.”

An inverse trend for the Warmth subscale emerged, where the virtual robot was rated slightly higher. This counter-intuitive finding invites several interpretations, which must be treated as hypotheses given the preliminary nature of the data. Based on the results obtained so far, we have already validated that the virtual robot serves as a reliable proxy for emotion recognition tasks. Therefore, the slightly higher Warmth scores are unlikely to stem from differences in embodiment itself, but rather may be attributed to the specific content of the conversations. This hypothesis should be rigorously tested with a statistically significant participant sample to assess its consistency. One other possible explanation is that users may feel less inhibited and more open when interacting with the virtual robot. Its screen-based format might create a less intimidating presence, thereby facilitating more comfortable and emotionally expressive dialogue.

It is crucial to heavily qualify these interpretations. Given the pilot study's limited sample size, these observations are merely indicative trends. They do not offer statistical proof but rather formulate a strong hypothesis for future validation.

Therefore, future research must rigorously test this trend with a larger participant pool. Such studies are essential to analytically confirm these preliminary observations. Furthermore, as suggested by Bertacchini et al. [6], future work should investigate how the addition of synchronized speech clarifies the meaning of even the most ambiguous gestures from our main experiment, creating a truly multimodal and context-aware empathetic interaction. Even the least recognizable gestures may become clearer when contextualized with verbal exclamations.

Chapter 6

Conclusions and future work

This thesis investigated a central question in Human-Robot Interaction: does the physical embodiment of a social robot alter how humans perceive its emotional expressions compared to a visually and kinematically identical digital twin? The core hypothesis was that the tangible presence of the NAO robot would lead to higher emotion recognition accuracy. To test this, a meticulous development process was undertaken to create a high-fidelity virtual replica of the NAO robot and a unified software architecture capable of deploying the exact same emotional gestures on both platforms, thereby isolating embodiment as the sole independent variable.

To achieve this goal, we designed and developed a comprehensive experimental framework. The cornerstone of this work was the creation of a meticulous digital twin, ensuring 1:1 visual and kinematic parity with the physical NAO robot. This involved a development pipeline from 3D modeling in Blender to a fully articulable, real-time controlled agent in Unity. A unified software architecture was engineered to deploy a set of 12 emotional gestures, derived from HRI literature, to both the physical and virtual robots using identical command signals.

A within-subjects experiment was conducted where participants identified emotions expressed by both the physical and virtual robot. The primary conclusion drawn from the results is a compelling null effect, with no statistically significant difference found in emotion recognition accuracy between the two conditions. Rather than embodiment, the analysis revealed the most powerful predictor of success was the intrinsic clarity and legibility of the emotional gestures themselves. This finding is significant because it provides strong empirical evidence that for research focused on non-verbal cues, a high-fidelity digital twin can serve as a valid, scalable, and cost-effective substitute for a physical robot, thereby opening avenues for more accessible and rapid prototyping in HRI.

Contributing directly to the ongoing HRI debate, this work provides a crucial qualification to the idea of embodiment's importance: while physical presence may be critical for some interaction types, it is not a prerequisite for conveying non-verbal emotional information. However, qualitative feedback hinted at an experiential difference, with some users finding the physical robot more “comfortable”, suggesting that functional equivalence—both platforms delivering the same objective emotion-recognition accuracy—does not automatically guarantee experiential equivalence. This distinction between functional and experiential perception paved the way for the second part of this work.

The preliminary assessment of the LLM-integrated framework, developed as a secondary objective of this thesis, has opened up new and fascinating avenues for research. The pilot study revealed a complex and counter-intuitive set of user perceptions. Participants tended to rate the physical robot as significantly more competent and capable, despite it being driven by the exact same LLM logic as its virtual counterpart. This suggests that physical embodiment confers a psychological "aura of capability" that transcends the agent’s actual programming. At the same time, the virtual robot was perceived as slightly warmer and more emotional, when the study we have conducted so far has validated equality in expressing emotions for both counterparts. These preliminary findings are a powerful testament to the nuanced effects of embodiment in more complex, conversational contexts.

The hypothesis that has emerged during this preliminary study is that despite parity for some tasks, the physical robot is perceived as more “comfortable” for long-term complex interactions.

Building on these findings, future work will integrate the empathetic Large Language Model (LLM) framework developed as a secondary objective of this thesis. The next research phase will move beyond static, non-verbal expressions to explore a constrained, multimodal interaction. For example a scenario where a user plays a logic game, such as chess, against the NAO robot. In this context, the LLM will not only manage the game logic and conversational turns but also trigger contextually appropriate emotional expressions. For instance, the robot might express 'happiness' after a successful move, 'sadness' upon losing a piece, or 'fear' when its king is in check. This creates a richer interaction where the primary research question will be to re-evaluate the role of embodiment in this more complex setting. Will the combination of speech, strategic dialogue, and emotional gestures makes the physical robot a more engaging, trustworthy, or effective game partner compared to its digital twin? thus exploring if the experiential comfort of physical presence translates into measurable behavioral outcomes.

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Chapter 7

Appendix

7.1 GitHub Repository for Code and Results

All scripts, source code, raw data, detailed analysis and supplementary results generated during this study are publicly available and can be fully accessed at the following GitHub repository:

`https://github.com/matteorigat/NAO`

7.2 Detailed Emotion Recognition Data

This appendix provides the complete, granular data from the experiment, complementing the summary statistics and analyses presented in Chapter 4 (Results). The following tables detail the performance for each of the 12 emotional stimuli, broken down by overall results, the physical robot condition, and the virtual robot condition.

These tables are provided to give a comprehensive view of participant responses, including not only recognition accuracy but also the dimensional ratings of valence and arousal, and the specific confusion patterns for each gesture.

Column Descriptions The table columns are defined as follows:

- **Emotion:** The name of the stimulus gesture (e.g., `Happiness1`).
- **Correct, Incorrect:** The number of correct and incorrect emotion identifications.
- **Correct_%:** Recognition accuracy, expressed as a percentage.

- **Valence_Mean, Valence_Var:** Mean and variance of perceived valence on a 9-point scale (1=unpleasant, 9=pleasant).
- **Arousal_Mean, Arousal_Var:** Mean and variance of perceived arousal on a 9-point scale (1=calm, 9=excited).
- **Frequent_Errors:** Most common misclassifications, with their frequency in parentheses.

	Emotion	Correct	Incorrect	Total	Correct_%	Valence_Mean	Valence_Var	Arousal_Mean	Arousal_Var	Frequent_Errors
6	Happiness1	24	0	24	100.0	8.21	1.04	5.17	6.41	-
8	Happiness3	24	0	24	100.0	7.83	0.75	4.79	6.61	-
3	Fear1	23	1	24	95.8	2.29	2.30	7.21	2.26	angry (1)
2	Anger3	21	3	24	87.5	3.00	2.35	6.38	2.42	fear (3)
0	Anger1	20	4	24	83.3	3.75	3.50	4.54	3.56	happy (2), sad (2)
4	Fear2	20	4	24	83.3	3.21	1.56	6.25	2.46	sad (2), angry (2)
9	Sadness1	19	5	24	79.2	2.75	2.11	4.08	2.34	angry (3), fear (2)
7	Happiness2	17	7	24	70.8	5.79	1.82	3.29	2.39	sad (3), angry (2), fear (2)
11	Sadness3	16	8	24	66.7	2.33	1.45	5.33	5.10	fear (6), angry (2)
5	Fear3	14	10	24	58.3	2.71	2.22	6.54	6.17	angry (6), sad (4)
1	Anger2	12	12	24	50.0	3.54	2.61	4.25	4.28	sad (8), happy (2), fear (2)
10	Sadness2	11	13	24	45.8	3.00	2.17	5.58	3.73	angry (7), fear (6)

Figure 20: Aggregated performance data for all 12 emotional stimuli across both physical and virtual conditions (N=24 participants, 288 total observations). Stimuli are ranked by their overall recognition accuracy. This table provides a holistic view of how effectively each gesture communicated its intended emotion to the entire participant pool.

	Emotion	Correct	Incorrect	Total	Correct_%	Valence_Mean	Valence_Var	Arousal_Mean	Arousal_Var	Frequent_Errors
3	Fear1	13	0	13	100.0	2.46	3.44	7.15	1.97	-
6	Happiness1	13	0	13	100.0	8.08	1.41	5.00	5.67	-
8	Happiness3	11	0	11	100.0	8.18	0.56	4.18	8.76	-
2	Anger3	11	2	13	84.6	3.23	2.86	6.23	2.36	fear (2)
9	Sadness1	8	2	10	80.0	2.80	4.40	3.80	2.18	angry (2)
4	Fear2	10	3	13	76.9	3.15	2.14	6.46	1.44	angry (2), sad (1)
1	Anger2	9	3	12	75.0	3.00	1.45	5.08	3.54	happy (1), fear (1), sad (1)
11	Sadness3	9	3	12	75.0	2.08	1.72	5.25	5.11	fear (2), angry (1)
0	Anger1	6	3	9	66.7	4.67	4.75	4.22	2.69	happy (2), sad (1)
5	Fear3	8	4	12	66.7	2.08	1.54	7.42	4.81	sad (2), angry (2)
7	Happiness2	8	5	13	61.5	6.00	2.83	2.69	1.40	angry (2), fear (2), sad (1)
10	Sadness2	6	7	13	46.2	3.46	2.27	5.15	2.47	angry (4), fear (3)

Figure 21: Detailed performance data for each emotional stimulus as expressed by the **physical NAO robot**. This table isolates the results from the "real" condition, showing accuracy, dimensional ratings, and error patterns when participants interacted with the tangible robot.

	Emotion	Correct	Incorrect	Total	Correct_%	Valence_Mean	Valence_Var	Arousal_Mean	Arousal_Var	Frequent_Errors
6	Happiness1	11	0	11	100.0	8.36	0.65	5.36	7.85	-
8	Happiness3	13	0	13	100.0	7.54	0.77	5.31	4.73	-
0	Anger1	14	1	15	93.3	3.20	2.17	4.73	4.21	sad (1)
2	Anger3	10	1	11	90.9	2.73	1.82	6.55	2.67	fear (1)
3	Fear1	10	1	11	90.9	2.09	1.09	7.27	2.82	angry (1)
4	Fear2	10	1	11	90.9	3.27	1.02	6.00	3.80	sad (1)
7	Happiness2	9	2	11	81.8	5.55	0.67	4.00	2.80	sad (2)
9	Sadness1	11	3	14	78.6	2.71	0.68	4.29	2.53	fear (2), angry (1)
11	Sadness3	7	5	12	58.3	2.58	1.17	5.42	5.54	fear (4), angry (1)
5	Fear3	6	6	12	50.0	3.33	2.24	5.67	6.42	angry (4), sad (2)
10	Sadness2	5	6	11	45.5	2.45	1.67	6.09	5.09	fear (3), angry (3)
1	Anger2	3	9	12	25.0	4.08	3.36	3.42	3.90	sad (7), happy (1), fear (1)

Figure 22: Detailed performance data for each emotional stimulus as expressed by the **virtual NAO robot (digital twin)**. This table isolates the results from the "virtual" condition, allowing for a direct comparison with the physical robot's performance shown in Figure 21.

7.3 Conversation Prompts for LLM Interaction

To facilitate a natural and engaging dialogue during the preliminary LLM assessment, participants were provided with a sheet of conversation prompts. This handout was designed to prevent stilted, factual exchanges and encourage more open-ended interactions where the robot could display a range of social and emotional behaviors. Figure 23 shows the content of the sheet.

Spunti di conversazione dialogo con Nao

Neutri

- Ti piacciono i film? Andiamo al cinema insieme?
- Andiamo al parco a prendere un gelato?
- Oggi fa proprio caldo, andiamo in un bar a prendere qualcosa di rinfrescante.
- Io adoro leggere i libri di avventura, quelli che parlano di isole tropicali.
- Dobbiamo prenotare le vacanze estive, dove ti piacerebbe andare al mare quest'anno?

Positivi

- Che bello, ho appena trovato 5€ per terra!
- Oggi è il mio compleanno, andiamo a festeggiare?
- Andiamo al parco a prendere un gelato? Oggi offro io, ho appena avuto la paghetta!
- Mi hanno appena regalato il nuovo Mario Kart, ci giochiamo assieme?
- Ho appena vinto 100€ alla lotteria, andiamo a fare shopping insieme!

Negativi

- Oh no, sono inciampato e mi sono fatto molto male.
- Andiamo al parco a prendere un gelato? Oh no, non ho soldi, offri tu oggi?
- Ho appena preso un brutto voto, cosa dico a mia mamma adesso?
- Cavolo, mi sono appena ricordato che non ho niente da mangiare stasera.
- Ho appena visto un ragno enorme!

Figure 23: The handout sheet with conversation prompts provided to participants to guide their dialogue with the LLM-powered NAO robot. The prompts were divided into neutral, positive, and negative categories to encourage a wide range of conversational topics.

7.4 Questionnaire Used in the Preliminary LLM Assessment

This section provides the Robotic Social Attributes Scale (ROSAS) questionnaire that was given to participants following their interaction with the LLM-powered robot (see Section 4.5). The form was used to gather subjective ratings on the robot's perceived social qualities. Figure 24 shows the exact layout of the questionnaire as presented to the participants.

Items (rate each one)	Not at all		Moderately		Very much
Competent	1	2	3	4	5
Capable	1	2	3	4	5
Responsible	1	2	3	4	5
Interactive	1	2	3	4	5
Reliable	1	2	3	4	5
Knowledgeable	1	2	3	4	5
Happy	1	2	3	4	5
Feeling	1	2	3	4	5
Social	1	2	3	4	5
Compassionate	1	2	3	4	5
Emotional	1	2	3	4	5
Organic	1	2	3	4	5
Scary	1	2	3	4	5
Strange	1	2	3	4	5
Awkward	1	2	3	4	5
Dangerous	1	2	3	4	5
Awful	1	2	3	4	5
Aggressive	1	2	3	4	5

Figure 24: The Robotic Social Attributes Scale (ROSAS) questionnaire used in the preliminary LLM assessment. Participants rated the robot on 18 adjectival items using a 5-point Likert scale, ranging from 1 ("Not at all") to 5 ("Very much"). This validated instrument measures user perceptions across three key dimensions: Competence, Warmth, and Discomfort.